

Options & Greeks 面试题库

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EN: 110 interview questions on options and Greeks for quant, valuation and market-risk roles. Market-sense questions come first. Every answer has an English version you can say out loud, then a Chinese version that explains the principle in more depth.

中文：共 110 道题，覆盖 quant、valuation、market risk 岗位最常考的 options 和 Greeks 问题。Market sense 题放在最前面。每道题先给英文答案（可以直接说出口），再给中文深度解析。

1. Market sense I：哪个期权更贵（Q1-Q12）

EN: There is no formula to plug into here; the interviewer is testing intuition. Use one frame every time: where is the forward, how much uncertainty is left ($\sigma\sqrt{T}$), and is there an extra right or restriction?

中文：这类题没有公式可套，考的是直觉。答题框架只有一句：**先问 forward 在哪里，再问剩下多少不确定性 ($\sigma\sqrt{T}$)，最后看有没有额外的权利或限制。**

Q1. Same strike, spot = strike: which is more expensive, the call or the put?

EN: The call, as long as rates exceed the dividend yield. Put-call parity gives $C - P = S - Ke^{-rT} > 0$. Intuitively, the forward sits above spot, so a spot-ATM call is in the money on a forward basis. They are equal only at the ATM-forward strike, $K = F$. For a high-dividend stock where $q > r$, the put is more expensive.

中文：利率高于分红率时 call 更贵。由平价关系， $C - P = S - Ke^{-rT} > 0$ 。直觉是 forward 高于 spot，所以 spot ATM 的 call 按 forward 看已经是 ITM。只有行权价等于 forward（ATM forward）时两者才一样贵。高分红股（ $q > r$ ）则是 put 更贵。

Q2. Is a 2-year option twice the price of a 1-year option?

EN: No. Variance adds linearly in time for independent returns, so the standard deviation of S_T grows like $\sigma\sqrt{T}$. An ATM-forward option is worth roughly the expected positive part of that distribution, about $0.4 \cdot S \cdot \sigma\sqrt{T}$. Doubling maturity multiplies the price by about $\sqrt{2} \approx 1.41$, not 2.

中文：不是。不重叠时段的收益相互独立，**方差可以相加，标准差不行**。两年的方差是 $2\sigma^2$ ，标准差是 $\sigma\sqrt{2}$ 。ATM 期权没有内在价值，价格只取决于到期分布有多宽。 $K = F$ 时 $d_1 = \frac{1}{2}\sigma\sqrt{T}$ 、 $d_2 = -\frac{1}{2}\sigma\sqrt{T}$ ，对 $N(\cdot)$ 在 0 处一阶展开即得：

$$C_{\text{ATMF}} = e^{-rT}F \cdot \left[2N\left(\frac{1}{2}\frac{\sigma\sqrt{T}}{\sqrt{2\pi}}\right) - 1 \right] \approx e^{-rT}F \cdot \frac{\sigma\sqrt{T}}{\sqrt{2\pi}} \approx 0.4 \cdot S \cdot \sigma\sqrt{T}$$

Case / 情形 ($S = 100$, $\sigma = 20\%$)	1Y	2Y	2Y / 1Y	Why / 原因
ATM, $r = 0$	7.97	11.25	1.41	Exactly $\sqrt{2}$ / 正好 $\sqrt{2}$
ATM forward, $r = 5\%$	7.97	11.25	1.41	Same as $r = 0$ / 与 $r = 0$ 相同
ATM spot, $r = 5\%$	10.45	16.13	1.54	Forward drifts away from K / forward 漂离行权价
ATM, $\sigma = 80\%$ (1Y vs 5Y)	31.08	62.89 (5Y)	2.02 ($\sqrt{5} = 2.24$)	$\sigma\sqrt{T}$ too large / 近似失效

EN (when it isn't $\sqrt{2}$): (1) With the strike fixed at spot, carry moves the forward away from K, adding a roughly linear-in-T intrinsic component. (2) When $\sigma\sqrt{T}$ is large the linear approximation fails and price grows slower than $\sqrt{T}$, because a call can never exceed the stock. (3) Implied vol differs by maturity, so compare total variance $\sigma^2(T) \cdot T$, which must be non-decreasing or there is calendar arbitrage.

中文 (什么时候不是 $\sqrt{2}$): ① 行权价固定在现价时, 期限越长 forward 离 K 越远, 多出一块大致随 T 线性增长的内在价值, 所以比例是 1.54。② $\sigma\sqrt{T}$ 很大时线性近似失效, 价格增长慢于 $\sqrt{T}$, 因为 call 永远不超过股价 ($\sigma = 80\%$ 时 25 年期 ATM call 为 95.5)。③ 各期限 implied vol 不同, 应比较总方差 $\sigma^2(T) \cdot T$, 它必须随 T 单调不减, 否则存在 calendar arbitrage。

Follow-up / 追问: Is a longer-dated option always more expensive?

EN: Time does two things: it adds variance, which always helps the holder, and it adds carry (strike discounting, dividends forgone), which can hurt. An American option can always mimic the shorter one, so longer is never cheaper. A European option cannot, so carry can win: a deep ITM European put or a high-dividend ITM European call can lose value with maturity.

中文: 时间有两个作用: 增加方差 (对多头永远有利) 和增加持有成本 (行权价折现、拿不到的分红, 可能不利)。美式期权可以随时行权, 长期至少能复制短期的结果, 不可能更便宜。欧式期权没有这个选择, carry 的不利影响超过方差的有利影响时, 长期反而更便宜。

Option / 期权	Longer always dearer? / 长期一定更贵?	Numbers / 数值
European call, no dividend / 无分红欧式 call	Yes / 是	See table above / 见上表
American call or put / 美式	Yes / 是	—

Option / 期权	Longer always dearer? / 长期限一定更贵？	Numbers / 数值
Deep ITM European put / 深度 ITM 欧式 put	No / 不一定	S = 60, K = 100, r = 5% : 0.5Y 37.5, 1Y 35.2, 2Y 31.2, 5Y 23.1
ITM European call, high dividend / 高分红 ITM 欧式 call	No / 不一定	S = 100, K = 60, q = 8% : 0.5Y 36.1, 1Y 32.4, 2Y 26.3, 5Y 15.1

EN: The deep ITM put is essentially a zero-coupon bond paying K, so receiving K later lowers its value; that is also why American puts get exercised early. The high-dividend call loses about 8% of forward per extra year, more than the added variance is worth.

中文：深度 ITM put 几乎肯定行权，相当于一份到期支付 K 的零息债，拿到 K 越晚现值越低，这也是美式 put 会被提前行权的原因。高分红 call 的持有人拿不到分红，每多一年 forward 就被削掉约 8%，超过了方差增加的价值。

Q3. American vs European: which is more expensive, and when are they equal?

EN: American $\geq$ European, because the early-exercise right can only add value. On a non-dividend stock the American call equals the European call, since early exercise is never optimal (Q28). The American put's early-exercise premium is largest when rates are high and the put is deep in the money.

中文：美式 $\geq$ 欧式，因为多了提前行权的权利。无分红的美式 call 等于欧式 call，因为提前行权永远不划算（Q28）。美式 put 的提前行权溢价在高利率、深度 ITM 时最大。

Q4. On an equity index, which is more expensive: the 25-delta put or the 25-delta call?

EN: The put, by several vol points. That is the negative skew (Q58): investors persistently buy downside protection, crashes are more likely than melt-ups, and vol rises when the market falls. In FX it depends on the sign of the risk reversal; safe-haven currency calls are usually richer.

中文：Put 更贵，高出几个 vol points。这就是负 skew（Q58）：投资者持续买下行保护，左尾的跳跃风险更大，跌市时波动率还会上升。FX 中哪一侧贵取决于 risk reversal 的符号，避险货币的 call 通常更贵。

Q5. Before or after an earnings announcement: when is a straddle more expensive?

EN: Before. Implied vol prices the known jump, then collapses after the release (vol crush). That's why you can buy a straddle, see a big move, and still lose money. You can back out the event vol from the term structure:

$$\sigma_{\text{total}}^2, T = \sigma_{\text{base}}^2, T + \sigma_{\text{event}}^2 \cdot \frac{1}{252}$$

中文：财报前更贵。隐含波动率会为已知的跳跃定价，公布后出现 **vol crush**。所以买了 straddle、股价也大幅变动，仍可能亏钱。从期限结构拆出事件波动率后，隐含的财报日涨跌幅大约是 $0.8 \times \sigma_{\text{event}}/\sqrt{252}$ ，系数 0.8 来自 Q82。

Q6. Which is cheaper: an Asian (average-price) call or a European call with the same strike?

EN: The Asian. An average moves less than the terminal price; with continuous arithmetic averaging the effective vol is about $\sigma/\sqrt{3}$, and the forward of the average is also lower. That's why corporates use Asians to hedge average costs cheaply.

中文：Asian 更便宜。平均价格的波动小于到期价格的波动：连续算术平均时，有效波动率约为 $\sigma/\sqrt{3}$ ，平均价格的 forward 也更低。所以公司客户用它以较低成本对冲平均采购成本。

Q7. Rank by price: vanilla call, up-and-out call, up-and-in call, lookback call.

EN: Lookback > vanilla > each barrier. Knock-in plus knock-out equals the vanilla, so neither barrier can exceed it. The lookback settles on the best price along the path, so it is the most expensive. The closer the barrier is to spot, the cheaper the knock-out and the closer the knock-in gets to the vanilla.

中文：Lookback > vanilla > 两个 barrier。Knock-in + knock-out = vanilla，所以任何一个 barrier 都不超过 vanilla。Lookback 按路径上最有利的价格结算，最贵。Barrier 越靠近现价，knock-out 越便宜，knock-in 越接近 vanilla。

Q8. Which is more expensive: a call on a basket, or a basket of calls?

EN: The basket of calls, or equal. With correlation below 1 the basket's vol is below the weighted average of the components' vols. Mathematically $\max(\Sigma x, 0) \leq \Sigma \max(x, 0)$, with equality only at perfect correlation.

中文：Basket of calls 更贵或相等。各资产不完全相关时，组合波动率小于各成分波动率的加权平均。从 payoff 看， $\max(\Sigma x, 0) \leq \Sigma \max(x, 0)$ （凸性），相关系数为 1 时才相等。

Q9. Why is index implied vol usually lower than the average single-stock implied vol?

EN: Diversification. Index variance depends on the correlations between constituents, so index vol embeds an implied correlation. A dispersion trade (sell index vol, buy single-stock vol) is a short-correlation trade. In a crisis correlations go to 1 and index vol catches up fast.

$$\sigma_{\text{index}}^2 = \sum_i w_i^2 \sigma_i^2 + \sum_{i \neq j} w_i w_j \rho_{ij} \sigma_i \sigma_j$$

中文：原因是分散化。指数方差取决于成分股之间的相关性，所以指数波动率隐含了市场对相关性的定价（implied correlation）。**Dispersion trade**（卖指数波动率、买成分股波动率）就是在卖相关性。危机时相关性趋向 1，指数波动率会快速追上。

Q10. Rates go up 100bp. What happens to calls and puts, and which are hit hardest?

EN: Calls rise and puts fall (Q64). Long-dated, deep ITM options move most because rho scales with $K \cdot T$; short-dated OTM options barely move. In practice a rate rise often comes with an equity sell-off, so the delta effect usually dominates rho.

中文：Call 涨、put 跌（Q64）。长期限、深度 ITM 的期权受影响最大，因为 $\rho \propto K\tau$ ；短期限 OTM 期权几乎不动。现实中加息往往伴随股市下跌，整体效果主要来自 delta 而不是 rho。

Q11. A company announces a special dividend. What happens to its calls and puts?

EN: The forward drops, so calls fall and puts rise. Deep ITM American call holders may exercise just before the ex-date. For listed options, exchanges often adjust the strike for special dividends but not for ordinary ones, which is a good practical point to raise.

中文：Forward 下降，call 跌、put 涨。深度 ITM 的美式 call 持有人可能在除息前提前行权。交易所上市的期权遇到特别分红时，交易所常会调整行权价，常规分红则不调整，这是很实务的加分点。

Q12. The market sells off 5% and implied vol jumps. You're long a 1-month 90% put: rank the P&L drivers.

EN: All three work for you. Delta: the put gains as spot falls. Gamma: as spot approaches the strike, delta becomes more negative, so gains accelerate. Vega: implied vol rises, and with negative skew low-strike vol rises most. Theta is a small cost. That's why OTM puts are crash insurance, and why they are expensive.

中文：三个驱动因子都对你有利。**Delta**：put delta 为负，标的下跌赚钱。**Gamma**：标的靠近行权价时 delta 更负，赚得越来越多。**Vega**：波动率上升，负 skew 下低行权价的 vol 涨得最多。Theta 是小成本。这就是 OTM put 是崩盘保险的原因，也是它平时贵的原因。

2. Market sense II：Greeks 如何随时间和行情变化（Q13-Q23）

EN: As expiry approaches, all the risk concentrates at the strike. ATM gamma and theta explode, ITM and OTM risk fades away, and vega decays at every strike.

中文：随着到期临近，所有风险都向行权价收缩：ATM 的 gamma 和 theta 急剧放大，ITM 和 OTM 的风险快速消失，vega 对所有行权价都在衰减。

Greek (long call)	ITM near expiry / 临近到期	ATM near expiry	OTM near expiry	Driver / 驱动原因
Delta	$\rightarrow 1$	Jumps between 0 and 1 / 在 0 与 1 间跳动	$\rightarrow 0$	Outcome becomes certain / 结果越来越确定
Gamma	$\rightarrow 0$	$\uparrow\uparrow, \propto 1/\sqrt{\tau}$	$\rightarrow 0$	Distribution narrows around spot / 分布变窄
Theta	Small (discounting only) / 很小	$\uparrow\uparrow$ in size, $\propto 1/\sqrt{\tau}$	Rises then falls to 0 / 先大后小	Time value $\propto \sqrt{\tau}$ / 时间价值 $\propto \sqrt{\tau}$
Vega	$\downarrow$	$\downarrow, \propto \sqrt{\tau}$	$\downarrow$	Remaining variance $\sigma^2\tau$ shrinks / 剩余方差变小

Q13. How does the time decay of an ATM option evolve as expiry approaches?

EN: It accelerates. ATM time value is about $0.4 \cdot S \cdot \sigma \sqrt{\tau}$, so theta scales like $1/\sqrt{\tau}$. The last quarter of an option's life takes half of its time value, since $\sqrt{1/4} = 1/2$. That's why option sellers like short-dated options: they collect the most theta, but also carry the most gamma.

中文：加速。ATM 时间价值 $\approx 0.4S\sigma\sqrt{\tau}$ ，求导得 $\Theta \propto 1/\sqrt{\tau}$ 。最后四分之一的存续期会损失一半的时间价值，因为 $\sqrt{1/4} = 1/2$ 。所以卖期权的人偏好短期限、收 theta，代价是承担最大的 gamma。

Q14. How does the theta of an OTM option evolve?

EN: Unlike ATM, OTM theta rises and then falls. Early on there is still meaningful time value to lose; near expiry the option is already worth almost nothing, so there is little left to decay. OTM decay peaks earlier than ATM decay.

中文：OTM 期权的 theta 先升后降。剩余时间较长时，它还有可观的时间价值可以流失；临近到期时价格已接近 0，没有什么可衰减的了。所以 OTM 的衰减峰值比 ATM 更早出现。

Q15. Spot hasn't moved for a week. What happened to the delta of your OTM call? Your ITM call?

EN: The OTM call's delta fell and the ITM call's delta rose. This is charm (Q68): with less time left, current moneyness decides the outcome more. Practically, your hedge drifts even when the market is flat, so you should compute weekend delta drift before Monday's open.

中文：OTM call 的 delta 下降，ITM call 的 delta 上升。这就是 charm (Q68)：剩余时间越少，当前的 moneyness 越决定最终结果。实务含义是，就算市场不动，对冲也会逐渐失效，周一开盘前要先算好周末的 delta 漂移。

Q16. Does an option lose three days of theta over a weekend?

EN: It depends on the time convention. Markets typically use volatility time, where weekends carry little weight because little information arrives. So weekend decay is priced in gradually before Friday's close rather than dropping all at once on Monday. If your model uses calendar days while the market uses trading days, Monday will show unexplained P&L.

中文：要区分日历时间和交易日时间。市场通常用**波动率时间**：周末几乎没有信息流，权重很低。所以周末的衰减在周五收盘前就已逐步体现在价格里，周一不会突然掉三天的价值。如果模型用日历日而市场用交易日，周一就会出现解释不了的 P&L，这对 valuation 和 P&L explain 很重要。

Q17. Spot rallies 10% through your long call's strike. What happens to each Greek?

EN: Starting ATM and ending ITM: delta rises toward 1, gamma falls, vega falls, and theta shrinks in size. On an equity index with negative skew, the rally also lowers implied vol, a small drag on a long call.

中文：假设涨前是 ATM、涨后变成 ITM：**Delta** 上升趋向 1；**Gamma** 下降，已远离 ATM；**Vega** 下降；**Theta** 绝对值变小。如果是负 skew 的股指，上涨还会拉低 implied vol，对 long call 是小幅不利因素。

Q18. Implied vol rises from 20% to 30%. What happens to the gamma of ATM vs OTM options?

EN: ATM gamma falls and OTM gamma rises. Since $\Gamma = \phi(d1)/(S\sigma\sqrt{\tau})$: at the money $\phi(d1)$ barely changes while σ in the denominator grows; out of the money higher vol pulls $d1$ toward 0, so $\phi(d1)$ rises a lot. Higher vol spreads gamma over a wider range of strikes. Longer maturity has the same effect.

中文：ATM gamma 下降，OTM gamma 上升。理由是 $\Gamma = \phi(d1)/(S\sigma\sqrt{\tau})$ ：ATM 时 $\phi(d1)$ 变化不大，分母里的 σ 变大，gamma 下降；OTM 时波动率上升使 $d1$ 更靠近 0， $\phi(d1)$ 大幅上升。直觉上，高波动率把 gamma “摊平”到更宽的行权价范围，“期限更长”也是同样的效果。

Q19. And the delta and theta after that vol rise?

EN: Delta is pulled toward 0.5 (Q44). ATM theta grows in size because $\Theta \approx -S\phi(d1)\sigma/(2\sqrt{\tau})$ is linear in σ . Higher vol means more rent per day, consistent with $\Theta \approx -\frac{1}{2}\sigma^2 S^2 \Gamma$ (Q52).

中文：Delta 向 0.5 靠拢 (Q44)。ATM 的 theta 绝对值变大，因为 $\Theta \approx -S\phi(d1)\sigma/(2\sqrt{\tau})$ 随 σ 线性增长。波动率越高，每天付的租金越多，与 $\Theta \approx -\frac{1}{2}\sigma^2 S^2 \Gamma$ (Q52) 一致。

Q20. It's expiry day. Which Greeks matter and which don't?

EN: What matters: gamma and pin risk near the strike, and ATM theta, which is now large per hour. What doesn't: vega ($\sqrt{\tau} \approx 0$) and rho. Delta can flip from 0 to 1 within a few ticks, so traders decide whether to close or roll positions early rather than hold into the close.

中文：很重要：ATM 附近的 gamma 和 pin risk，以及 ATM 每小时的 theta。几乎无关：vega ($\sqrt{\tau} \approx 0$) 和 rho。注意：delta 可能在几个 tick 内从 0 变到 1，交易员会决定要不要提前平仓或滚动到下个期限，而不是硬抗到收盘。

Q21. You're long a 3-month straddle. Spot doesn't move for 2 weeks. What happened?

EN: You lost money: theta bled every day with no realised moves to scalp (realised below implied, Q77). Quiet markets also tend to pull implied vol down, adding a vega loss. With less time left, gamma is now higher and vega lower, so the position drifts from a vega trade toward a gamma trade.

中文：亏钱。每天流失 theta，却没有实际波动可以 scalp（Q77 中的 $\sigma_{\text{realised}} < \sigma_{\text{implied}}$ ）。市场平静还往往让 implied vol 下降，所以还有 vega 损失。剩余期限变短后，gamma 变大、vega 变小，头寸从 vega 交易向 gamma 交易漂移。

Q22. Why do traders say “gamma is short-dated, vega is long-dated”?

EN: For ATM options $\Gamma \propto 1/\sqrt{\tau}$ and $\text{Vega} \propto \sqrt{\tau}$ (Q55). A 1-week option has about 7× the gamma of a 1-year option and about 1/7 of the vega ($\sqrt{52} \approx 7.2$). So you trade realised vol with short-dated options and the level of implied vol with long-dated ones.

中文：ATM 的 $\Gamma \propto 1/\sqrt{\tau}$ 、 $\text{Vega} \propto \sqrt{\tau}$ (Q55)。同样数量的期权，1周到期的 gamma 大约是 1 年到期的 7 倍，vega 却只有约 1/7 ($\sqrt{52} \approx 7.2$)。所以赌实际波动用短期限，赌隐含波动率水平用长期限。

Q23. How do option price and Greeks change before, at and after expiry?

EN: Before expiry, time value decays like $\sqrt{\tau}$, so decay accelerates; delta collapses toward a step function, and gamma and theta concentrate at the strike, blowing up like $1/\sqrt{\tau}$, while vega shrinks to zero. At expiry the option equals its payoff: delta is a step, gamma is a spike at K, and theta, vega and rho are zero. After expiry the risk transfers into a stock or cash position, so the main risks are pin risk and unwinding the hedge.

中文：以 $K = 100$ 、 $\sigma = 20\%$ 、 $r = 0$ 的 ATM long call 为例：

Time to expiry / 距到期	Price / 价格	Delta	Gamma	Theta per day / 每天	Vega per vol pt
3 months	3.99	0.52	0.040	-0.022	0.199
1 month	2.30	0.51	0.069	-0.038	0.115
1 week	1.10	0.51	0.144	-0.079	0.055
1 day	0.42	0.50	0.381	-0.209	0.021

- **到期前**：时间价值按 $\sqrt{\tau}$ 收缩，最后1周丢掉近 2/3。Delta 从 S 形曲线变成阶梯，比如 S = 95 时 1 个月 delta 0.20，1 周只剩 0.03。ATM gamma 和 theta 在 1 天时是 3 个月时的约 10 倍；vega 按 $\sqrt{\tau}$ 缩小。
- **到期那一刻**：价格等于 payoff；delta 是阶梯（ITM 为 1、OTM 为 0）；gamma 是 K 处的尖峰；theta、vega、rho 全为 0。
- **到期之后**：实物交割时，ITM long call 变成股票头寸，原先卖出的对冲要么抵消要么需要平仓；现金交割时只剩现金，对冲必须平掉。还要注意 **pin risk**（不确定对手是否行权，周末跳空无法对冲）和**结算价风险**（比如 SPX 按开盘价结算）。

Greeks 速查表 / Cheat sheet

EN: Black-Scholes formulas and signs for a **long** position, no dividends. ϕ is the standard normal density, N the standard normal CDF, $\tau = T - t$.

中文：以下是 Black-Scholes 下、无分红、**long** 头寸的公式和符号。 ϕ 是标准正态密度，N 是标准正态累积分布函数， $\tau = T - t$ 。

Greek	Definition / 定义	Call	Put	Sign, long call / put 符号	Largest where / 最大值出现在
Delta Δ	$\partial V / \partial S$	$N(d1)$	$N(d1) - 1$	+ / -	Deep ITM / 深度 ITM (绝对值趋近 1)
Gamma Γ	$\partial^2 V / \partial S^2$	$\phi(d1) / (S\sigma\sqrt{\tau})$	Same as call / 同 call	+ / +	ATM, near expiry / ATM、临近到期
Vega ν	$\partial V / \partial \sigma$	$S \phi(d1) \sqrt{\tau}$	Same as call / 同 call	+ / +	ATM, long-dated / ATM、长期限
Theta Θ	$\partial V / \partial t$	$-S\phi(d1)\sigma/(2\sqrt{\tau}) - rK e^{(-r\tau)} N(d2)$	$-S\phi(d1)\sigma/(2\sqrt{\tau}) + rK e^{(-r\tau)} N(-d2)$	Usually - / - 通常为负	ATM, near expiry / ATM、临近到期
Rho ρ	$\partial V / \partial r$	$K\tau e^{(-r\tau)} N(d2)$	$-K\tau e^{(-r\tau)} N(-d2)$	+ / -	Deep ITM, long-dated / 深度 ITM、长期限
Vanna	$\partial^2 V / \partial S \partial \sigma$	$-\phi(d1) d2 / \sigma$	Same as call / 同 call	Depends on moneyness / 视 moneyness 而定	Near 25-delta / 25-delta 附近
Volga	$\partial^2 V / \partial \sigma^2$	$Vega \cdot d1 d2 / \sigma$	Same as call / 同 call	+ in wings, ≈ 0 ATM / wings 为正, ATM 约为 0	Wings

$$d_1 = \frac{\ln(S/K) + (r - q + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \\ \text{and } d_2 = d_1 - \sigma\sqrt{\tau}$$

EN: The table's vega is per 1.00 change in σ . Desks usually quote vega **per vol point** (σ moves by 0.01, divide by 100) and theta **per day** (divide by 365 or 252). State your units first in an interview; it sounds professional.

中文：表里的 Vega 按 σ 变动 1.00 计。实务中一般报 **per vol point** (σ 变动 0.01, 除以 100) ; Theta 一般报 **per day** (除以 365 或 252)。面试时先说清楚单位, 会显得很专业。

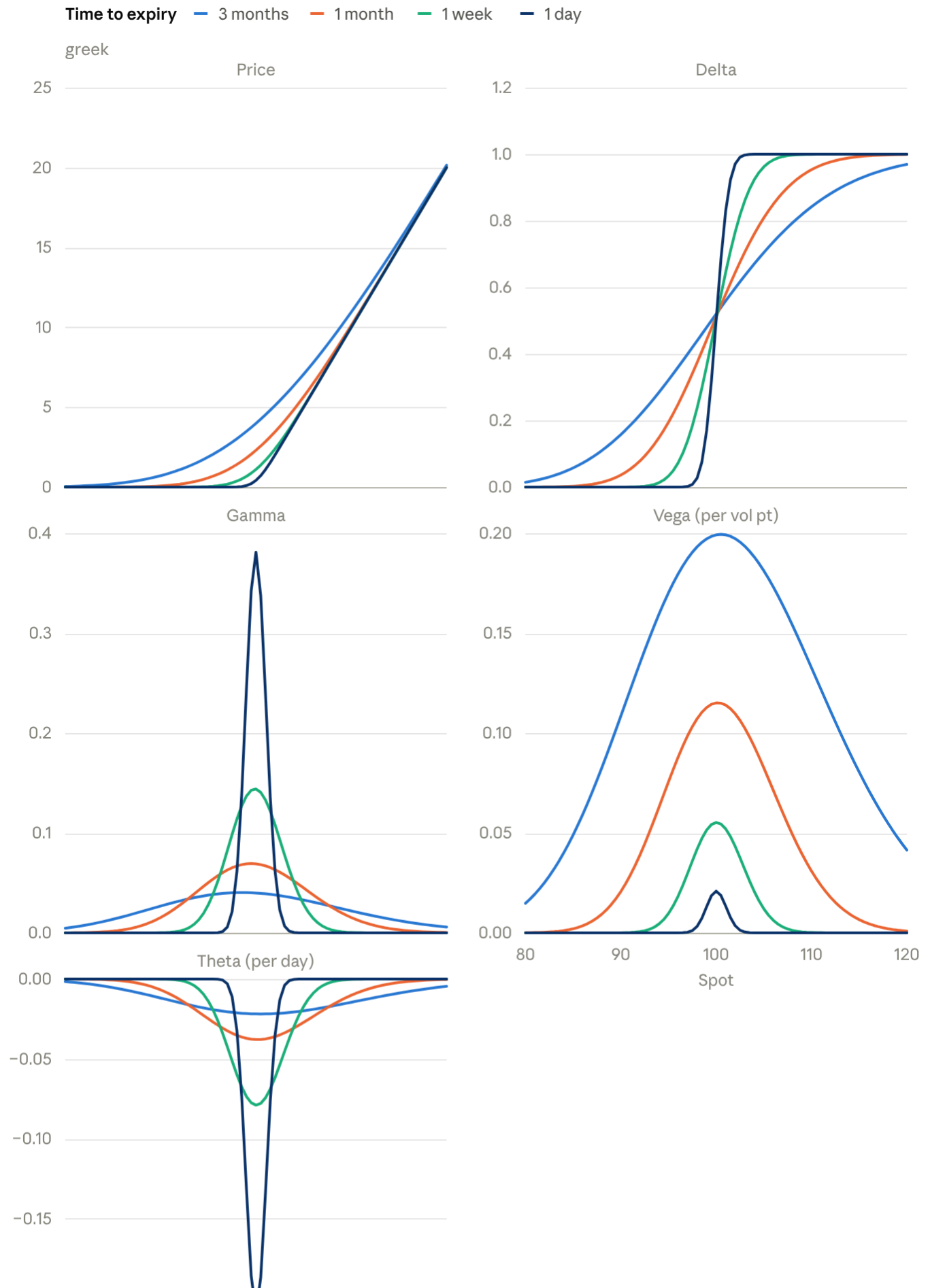
Greeks 曲线图 / Greek profiles

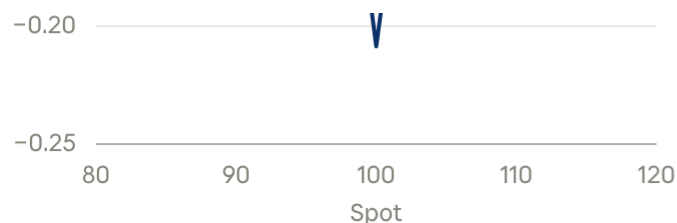
EN: Three pictures cover almost every “how does this Greek change when...” question: Greeks against spot, against time to expiry, and against volatility. All are for a long call with $K = 100$ and $r = 0$; put gamma, vega and theta are identical, and put delta is call delta minus 1.

中文：面试里“某个 Greek 随什么怎么变”的题, 几乎都能用这三张图回答: Greeks 对 spot、对剩余期限、对波动率。均为 $K = 100$ 、 $r = 0$ 的 long call ; put 的 gamma、vega、theta 与 call 完全相同, put delta = call delta - 1。

1. Greeks vs spot / 随标的价格变化

Long call Greeks vs spot





K = 100, vol 20%, r = 0; one line per time to expiry; vega per vol point, theta per day
 Black-Scholes, computed in Python · K = 100, $\sigma = 20\%$, spot 80–120

EN:

- **Price** is convex in spot and collapses onto the hockey-stick payoff as expiry nears; the gap between the curve and the payoff is time value, largest at the strike.
- **Delta** is the slope of the price curve: an S-shape that sharpens into a step at K as time runs out.
- **Gamma** is the slope of delta, a bell centred near the strike. Its area across all spots always equals 1 (delta goes from 0 to 1), so as the bell gets narrower near expiry it must get taller: that's why short-dated ATM gamma explodes.
- **Vega** is also a bell near the strike, but it shrinks with $\sqrt{\tau}$, so short-dated options have little vega.
- **Theta** is the mirror image of gamma, because $\Theta \approx -\frac{1}{2}\sigma^2 S^2 \Gamma$: wherever you own the most convexity you pay the most rent.

中文：

- **Price**：关于 spot 是凸函数，临近到期时贴向“曲棍球杆”形的 payoff。曲线与 payoff 之间的距离就是时间价值，在行权价处最大。
- **Delta**：价格曲线的斜率，是一条 S 形曲线，越接近到期越陡，最后变成 K 处的阶梯。
- **Gamma**：delta 的斜率，是行权价附近的钟形曲线。关键原理：**gamma 曲线下的面积恒等于 1**（delta 从 0 变到 1），所以临近到期钟形变窄时必然变高，这就是短期 ATM gamma 爆炸的原因。
- **Vega**：同样是行权价附近的钟形，但整体按 $\sqrt{\tau}$ 缩小，所以短期限期权几乎没有 vega。
- **Theta**：gamma 的镜像，因为 $\Theta \approx -\frac{1}{2}\sigma^2 S^2 \Gamma$ ：你持有凸性最多的地方，付的租金也最多。

2. Greeks vs time to expiry / 随剩余期限变化

Black-Scholes, computed in Python · 1 year to 1 day

EN:

- **Delta**: the ITM call heads to 1 and the OTM call to 0, while the ATM call stays near 0.5; with little time left, today's moneyness decides the outcome (charm).

- **Gamma and theta:** ATM both grow like $1/\sqrt{\tau}$ and explode in the last days. ITM and OTM gamma and theta first hold up, then collapse to zero once the strike is several standard deviations ($\sigma\sqrt{\tau}$) away.
- **Vega** falls for every strike. ATM vega decays like $\sqrt{\tau}$; ITM and OTM vega die much faster, because a vol change can no longer move them across the strike.
- The key quantity is $\sigma\sqrt{\tau}$, the remaining standard deviation: moneyness measured in those units decides whether an option still behaves like an option.

中文：

- **Delta**：ITM 趋向 1，OTM 趋向 0，ATM 始终在 0.5 附近。剩余时间越少，当前 moneyness 越决定结果（charm）。
- **Gamma 和 theta**：ATM 的两者都按 $1/\sqrt{\tau}$ 增长，最后几天爆炸。ITM 和 OTM 的先维持一段，等行权价距现价超过几个标准差（ $\sigma\sqrt{\tau}$ ）后就塌缩到 0。
- **Vega**：所有行权价都下降。ATM 按 $\sqrt{\tau}$ 衰减；ITM 和 OTM 衰减得快得多，因为波动率变化已经无法把它们推过行权价。
- 核心量是 $\sigma\sqrt{\tau}$ （**剩余标准差**）：用它为单位衡量的 moneyness，决定一个期权还像不像期权。

3. Greeks vs volatility / 随波动率变化

Black-Scholes, computed in Python · σ 5%–60%, 3 months

EN:

- **Price** rises with vol for every strike. The ATM line is almost straight (vega nearly constant, volga ≈ 0), while OTM and ITM lines curve upward (positive volga), which is why wings trade at a vol premium.
- **Delta** is pulled toward 0.5: OTM delta rises and ITM delta falls, because higher vol makes today's moneyness less decisive (vanna).
- **Gamma** at the money falls like $1/\sigma$; away from the money it first rises, peaks when the strike is about one standard deviation away, then falls. Higher vol spreads gamma across a wider range of spots.
- **Vega** at the money is nearly flat in vol; OTM and ITM vega rise with vol, the same positive volga seen in the price panel.
- Raising vol has the same effect as lengthening maturity, because both only enter through $\sigma\sqrt{\tau}$.

中文：

- **Price**：所有行权价都随波动率上升。ATM 几乎是直线（vega 近乎常数、volga ≈ 0 ），OTM 和 ITM 向上弯（volga 为正），这就是 wings 有波动率溢价的原因。
- **Delta**：向 0.5 靠拢，OTM 上升、ITM 下降（vanna）。
- **Gamma**：ATM 按 $1/\sigma$ 下降；远离 ATM 时先升后降，在行权价大约离现价一个标准差时达到峰值。高波动率把 gamma 摊平到更宽的价格范围。
- **Vega**：ATM 几乎不随波动率变；OTM 和 ITM 随波动率上升，就是价格面板上那个正 volga。
- 提高波动率和拉长期限效果相同，因为两者只通过 $\sigma\sqrt{\tau}$ 进入公式。

3. 期权基础 / Option basics (Q24-Q32)

Q24. What is put-call parity, and how do you prove it?

$$C - P = S e^{-q\tau} - K e^{-r\tau}$$

EN: Long a call and short a put with the same strike and expiry pays $S_T - K$ at expiry, exactly like a forward struck at K . Identical payoffs must have identical prices today, otherwise you buy the cheap one and sell the dear one. It is model-free and holds for European options only. Consequences: calls and puts on the same strike must share the same implied vol, $\Delta_{\text{call}} - \Delta_{\text{put}} = 1$ (no dividends), and gamma and vega are identical.

中文：证明用无套利。long call + short put 的到期 payoff 恒等于 $S_T - K$ ，正好是一份行权价为 K 的 forward。两个组合到期 payoff 完全相同，今天的价格就必须相同，否则可以买便宜的、卖贵的。它**不依赖任何模型**，只适用于欧式期权。推论：同一行权价的 call 和 put implied vol 必须相同； $\Delta_{\text{call}} - \Delta_{\text{put}} = 1$ （无分红）；gamma、vega 完全相同。

Q25. What are intrinsic value and time value?

EN: Intrinsic value is what you'd get by exercising now, e.g. $\max(S - K, 0)$ for a call. Time value is the option price minus intrinsic value; it comes from the uncertainty left before expiry. It peaks at the money, where the final outcome is most uncertain, and goes to zero deep in or out of the money.

中文：Intrinsic value 是立即行权的价值，比如 call 是 $\max(S - K, 0)$ 。Time value 是期权价格减去内在价值，来自剩余时间内的**不确定性**（optionality）。它在 ATM 最大，因为最终落在 ITM 还是 OTM 最不确定；深度 ITM 或 OTM 时结果几乎已定，时间价值趋近 0。

Q26. Define moneyness. How do traders quote it?

EN: The simplest measure is K/S or K/F . Traders prefer delta (a 25-delta put, a 10-delta call) because delta accounts for time and vol: $K = 110$ is deep OTM for a 1-week option but nearly ATM for a 5-year one. FX vol surfaces are quoted by delta: ATM, 25 Δ RR/BF and 10 Δ RR/BF.

中文：最简单的定义是 K/S 或 K/F 。交易上更常用 **delta** 表示 moneyness（如 25-delta put、10-delta call），因为 delta 同时考虑了时间和波动率：同样是 $K = 110$ ，对 1 周到期是深度 OTM，对 5 年到期差不多是 ATM。FX 市场的 vol surface 就是按 delta 报价的。

Q27. What are the no-arbitrage bounds on a European call?

$$\max(S e^{-q\tau} - K e^{-r\tau}, 0) \leq C \leq S e^{-q\tau}$$

EN: Upper bound: a call can't be worth more than the stock, because the stock always pays at least as much. Lower bound: from put-call parity and $P \geq 0$. Also, call prices must be decreasing and convex in the strike; otherwise a call spread or a butterfly is an arbitrage.

中文：上界：call 不可能比股票本身更贵，因为持有股票的 payoff 总是更高。下界：由平价关系且 $P \geq 0$ 推出。此外 call price 对 K 必须**单调递减且凸**，否则可以构造 call spread 或 butterfly 套利。

Q28. Would you ever exercise an American call on a non-dividend stock early?

EN: Never. From the lower bound, $C \geq S - K e^{-(r\tau)} > S - K$, so the live option is always worth more than exercising. Early exercise throws away two things: interest on the strike you'd pay early, and the insurance value against a fall. Keep it or sell it instead.

中文：不会。按 Q27 的下界， $C \geq S - K e^{-(r\tau)} > S - K$ ，活着的期权永远比行权更值钱。提前行权会失去两样东西：行权价推迟支付的利息，以及股价下跌时的保护（时间价值）。所以要么继续持有，要么直接卖掉。

Q29. When is early exercise optimal?

EN: An American call: only just before an ex-dividend date, when the dividend exceeds the remaining time value plus interest on the strike. An American put: when it is deep in the money and rates are high, because exercising collects K now and earns interest, while the stock can fall at most to zero. American puts therefore have no closed form and need a binomial tree, a PDE, or Longstaff-Schwartz regression Monte Carlo.

中文：美式 call 只可能在**除息日前一刻**行权，条件是分红大于剩余时间价值加行权价的利息。美式 put 在深度 ITM 且利率较高时可能提前行权：行权能马上拿到 K 、早收利息，而股价最多跌到 0，继续持有的上行空间有限。所以美式 put 没有闭式解，需要用 binomial tree、PDE 或 Longstaff-Schwartz。

Q30. Why does a call option price increase with volatility, even though volatility is “bad”?

EN: Because the payoff is convex and asymmetric: unlimited upside, downside capped at the premium. Higher vol fattens both tails, but the holder keeps the good tail and the bad tail is truncated. That is Jensen's inequality: for convex f , $E[f(S)] \geq f(E[S])$.

中文：因为 payoff 是凸的、不对称的：上行无限，下行最多损失期权费。波动率越大，尾部越厚，正尾的收益增加，负尾的损失却被截断。这就是 Jensen 不等式：对凸函数 f , $E[f(S)] \geq f(E[S])$ 。

Q31. What is a forward vs a futures, and how does it relate to options?

EN: Both lock in a future price. A forward is OTC and settles at maturity; a future is exchange-traded and marked to market daily. When rates and the underlying are positively correlated, futures trade slightly above forwards (convexity adjustment). Option pricing is really forward-based: Black-76 replaces S with $F = Se^{((r-q)\tau)}$, so the ATM-forward strike is the true point of symmetry.

中文：两者都是在未来以约定价格成交。forward 是 OTC、到期结算；futures 在交易所交易、每日盯市。利率与标的正相关时，futures 价格略高于 forward (convexity adjustment)。期权定价其实是基于 forward 的：Black-76 把 S 换成 F ，所以 **ATM forward** 才是真正的对称点。

Q32. Explain common option strategies and what view each expresses.

Strategy / 策略	Structure / 结构	View / 观点	Main Greeks
Straddle	Long ATM call + long ATM put	Big move, direction unknown / 大幅波动、方向不定	$\Delta \approx 0$, $\Gamma +$, Vega+, $\Theta -$
Strangle	Long OTM call + long OTM put	Same but cheaper, needs a bigger move / 更便宜但需要更大波动	$\Gamma +$, Vega+
Bull call spread	Long call K_1 + short call K_2 ($K_1 < K_2$)	Mildly bullish, cheaper / 温和看涨	$\Delta +$, small vega
Risk reversal	Long OTM call + short OTM put	Bullish, trades skew / 看涨并报价 skew	$\Delta +$, Vanna
Butterfly	Long K_1 and K_3 , short $2 \times K_2$	Low vol, pin at K_2 / 低波动、收在 K_2	$\Gamma -$ at body, Volga
Covered call	Long stock + short call	Mildly bullish, earn premium / 温和看涨、赚期权费	$\Delta < 1$, $\Gamma -$

EN: A butterfly's price divided by the squared strike spacing approximates the risk-neutral density (Breedon-Litzenberger), a common follow-up.

中文：Butterfly 的价格除以行权价间隔的平方，就是风险中性密度的近似 (Breedon-Litzenberger)，这是高频追问。

4. Black-Scholes 原理 / Principles (Q33-Q40)

Q33. What are the assumptions of Black-Scholes? Which ones fail most in practice?

EN: GBM with lognormal prices, constant volatility and rates, continuous frictionless trading and hedging, no short-sale limits, no arbitrage, no jumps. The ones that fail most: constant vol (we see smiles, skews and term structures), no jumps (earnings, crashes), and continuous hedging (hedging is discrete and costly). These failures motivated local vol, stochastic vol (Heston, SABR) and jump-diffusion (Merton 1976).

中文：假设包括：标的服从 GBM（对数正态）；波动率和利率为常数；可以连续交易并对冲；没有交易成本和卖空限制；没有套利；价格没有跳跃。最常失效的是：**常数波动率**（现实有 smile、skew 和期限结构）、**没有跳跃**（财报、崩盘）、**连续对冲**（只能离散对冲且有成本）。这些失效催生了 local vol、stochastic vol（Heston、SABR）和 jump diffusion（Merton 1976）。

Q34. Derive the Black-Scholes PDE.

EN: Form $\Pi = V - \Delta S$ and apply Itô's lemma to V . Choosing $\Delta = \partial V / \partial S$ cancels the dW term, so the portfolio is riskless. A riskless portfolio must earn the risk-free rate, $d\Pi = r\Pi dt$, which gives the PDE. The key point is that μ drops out.

$$dV = \left(\frac{\partial V}{\partial t} + \mu S \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt + \sigma S \frac{\partial V}{\partial S} dW$$

$$\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + r S \frac{\partial V}{\partial S} - rV = 0$$

中文：构造组合 $\Pi = V - \Delta S$ ，对 V 用 Itô 引理。取 $\Delta = \partial V / \partial S$ ， dW 项被消掉，组合变成无风险。无风险组合只能赚无风险利率，即 $d\Pi = r\Pi dt$ ，由此得到 PDE。关键是 **μ 消失了**，所以期权价格与投资者对期望收益的看法无关。

Q35. Why doesn't the expected return μ appear in the price?

EN: Because the option can be replicated by dynamic hedging, its price is the cost of the hedge, which depends only on today's stock price. That price already reflects the market's view of μ and risk preferences. Risk-neutral pricing is a computational device, not a claim that investors are risk-neutral.

中文：因为期权可以用标的加现金**完全复制**。复制成本只取决于标的今天的价格，而今天的价格已经反映了市场对 μ 和风险偏好的共识。对冲消除了方向风险，所以可以在风险中性测度下定价。风险中性只是计算工具，并不表示投资者真的风险中性。

Q36. What do $N(d1)$ and $N(d2)$ mean?

EN: $N(d_2)$ is the risk-neutral probability of finishing in the money, $Q(S_T > K)$, which is why a cash-or-nothing call is worth $e^{-r\tau}N(d_2)$. $N(d_1)$ is the call delta, and also the in-the-money probability under the measure that uses the stock as numeraire; $S \cdot N(d_1)$ is the asset-or-nothing call. So a call = asset-or-nothing - $K \times$ cash-or-nothing. Saying “delta is the probability of finishing ITM” is only an approximation, since $N(d_2) < N(d_1)$.

中文： $N(d_2)$ 是风险中性测度下到期落在 ITM 的概率 $Q(S_T > K)$ ，所以 cash-or-nothing 的价格是 $e^{-r\tau}N(d_2)$ 。 $N(d_1)$ 是 call 的 delta，也是以股票为 numeraire 的测度下 ITM 的概率； $S \cdot N(d_1)$ 是 asset-or-nothing call。所以 call = asset-or-nothing - $K \times$ cash-or-nothing。常见陷阱是说“delta 就是 ITM 概率”，它只是近似。

Q37. What is risk-neutral pricing and the martingale measure?

EN: The first fundamental theorem: no arbitrage if and only if there is an equivalent measure Q under which all asset prices divided by the numeraire (usually the bank account) are martingales. The second: Q is unique if and only if the market is complete. The option price is the discounted Q -expectation of the payoff. Girsanov's theorem moves from P to Q by changing the drift from μ to r while leaving vol unchanged.

$$V_0 = e^{-rT} \mathbb{E}^Q[\text{payoff}(S_T)]$$

中文：第一基本定理：市场无套利，当且仅当存在等价鞅测度 Q ，使所有资产价格除以 numeraire（通常是银行账户）后都是鞅。第二基本定理： Q 唯一，当且仅当市场完备。期权价格就是 payoff 在 Q 下的折现期望。从 P 到 Q 由 Girsanov 定理完成：漂移项从 μ 换成 r ，波动率不变。

Q38. Why is volatility the only unobservable input?

EN: S , K , T , r and dividends are observable from the contract or the market; future volatility isn't. So markets quote options in implied vol, and BS becomes “the wrong number in the wrong formula to get the right price”: a translator that makes prices comparable across strikes and maturities.

中文： S 、 K 、 T 、 r 、分红都能从合约或市场观察到，只有未来的波动率不行。所以市场直接用 implied vol 报价，BS 公式被当成“把价格翻译成可以横向比较的单位”的工具。

Q39. How would you price an option with Monte Carlo vs a PDE vs a tree? When would you use each?

Method / 方法	Good for / 适合	Poor for / 不适合	Convergence / 收敛
Monte Carlo	Path-dependent (Asian, lookback), many assets / 路径依赖、高维	American (needs LSM), noisy Greeks / 美式、Greeks 噪声大	Error $\sim 1/\sqrt{N}$, dimension-free

Method / 方法	Good for / 适合	Poor for / 不适合	Convergence / 收敛
PDE / finite differences	Low dimension (≤ 3), American, barriers / 低维、美式、障碍	High dimension / 高维	Crank-Nicolson $O(\Delta t^2, \Delta S^2)$
Binomial tree	American, teaching, quick checks / 美式、快速检验	Path-dependent, high dimension	Error $\sim 1/n$, oscillates / 有振荡
Closed form / 闭式解	European vanilla, digitals, some barriers	Most exotics	Exact

EN: From the CQF project: under GBM I'd simulate S_T exactly, and only time-step for path-dependent payoffs or models without a closed-form transition density.

中文：结合你的 CQF 项目可以补充：GBM 下可以直接精确模拟 S_T ，只有路径依赖的 payoff 或没有闭式转移密度的模型才需要分步模拟。

Q40. What is the difference between Euler-Maruyama and Milstein?

EN: Euler keeps only the first-order Itô-Taylor terms: strong order 0.5, weak order 1.

Milstein adds $\frac{1}{2} b b' (\Delta W^2 - \Delta t)$, lifting strong order to 1 while weak order stays 1. So for a smooth payoff like a European call Milstein doesn't improve the price; for digitals and barriers, which depend on which side of a level the path ends, it helps a lot. My notebook has the numbers.

中文：Euler 只保留 Itô-Taylor 展开的一阶项，强收敛阶 0.5、弱收敛阶 1。Milstein 多加一项 $\frac{1}{2} b b' (\Delta W^2 - \Delta t)$ ，强收敛阶提高到 1，弱收敛阶仍是 1。所以对 European call 这类光滑 payoff，Milstein 不提高定价精度；对 digital、障碍这类依赖路径是否越过某水平的 payoff，Milstein 明显更好。你的 notebook 里有现成的数字可以引用。

5. Delta (Q41-Q48)

Q41. What is delta, and what is the delta of an ATM call?

EN: Delta is the first-order sensitivity of the option price to the underlying, i.e. the hedge ratio: $[0, 1]$ for calls, $[-1, 0]$ for puts. A spot-ATM call has delta slightly above 0.5, because at $K = S$, $d1 = (r + \frac{1}{2}\sigma^2)\sqrt{\tau}/\sigma > 0$. With $\sigma = 20\%$, $r = 5\%$, $T = 1$, $d1 = 0.35$ and $\text{delta} \approx 0.64$. The strike with exactly 0.5 delta (the delta-neutral straddle strike) sits slightly above the forward.

中文：Delta 是期权价格对标的价格的一阶敏感度，也就是对冲比例。Call 在 $[0, 1]$ ，put 在 $[-1, 0]$ 。Spot ATM call 的 delta **略大于 0.5**，因为 $K = S$ 时 $d1 > 0$ 。以 $\sigma = 20\%$ 、 $r = 5\%$ 、 $T = 1$ 为例， $d1 = 0.35$ ， $\text{delta} \approx 0.64$ 。波动率越高、期限越长，ATM delta 越偏离 0.5。delta 恰好等于

0.5 的行权价 (DNS) 略高于 forward。

Q42. Is delta the probability of expiring in the money?

EN: Not exactly. The risk-neutral ITM probability is $N(d_2)$; delta is $N(d_1)$, which is larger, and the gap grows with $\sigma\sqrt{\tau}$. Both are risk-neutral, not real-world, probabilities. Traders use delta as a rough ITM probability, but say it's an approximation.

中文: 不完全是。风险中性下 ITM 的概率是 $N(d_2)$ ，delta 是 $N(d_1)$ ，前者更小，差异随 $\sigma\sqrt{\tau}$ 增大。而且两者都是 Q 测度下的概率，不是真实世界概率。交易员常把 delta 当成 ITM 概率的粗略估计，面试时要点明这是近似。

Q43. How does delta change as time to expiry decreases?

EN: It polarises toward 0 or 1: ITM goes to 1, OTM goes to 0, and near the money delta becomes a step function. That's why ATM gamma explodes before expiry. The rate of change of delta with time is charm (delta decay), which is why hedges need adjusting on Monday morning even with no market move.

中文: 临近到期时，delta 向 0 或 1 两极分化：ITM 趋向 1，OTM 趋向 0，ATM 附近变成阶梯函数。这就是 gamma 在到期前的 ATM 处暴涨的原因。Delta 随时间的变化率叫 **charm** (delta decay)，周一早上没有行情也要调整对冲，就是它造成的。

Q44. How does delta change when volatility increases?

EN: Higher vol pulls delta toward 0.5: OTM call delta rises and ITM call delta falls, because current moneyness matters less for the final outcome. This cross-sensitivity is vanna, $\partial\Delta/\partial\sigma = \partial\text{Vega}/\partial S$.

中文: 波动率上升把 delta 往 0.5 拉：OTM call 的 delta 上升，ITM call 的 delta 下降。原因是高波动下当前 moneyness 对最终结果的决定作用变小了。这个交叉敏感度就是 **vanna**。

Q45. What is the delta of a forward, a future, and the stock itself?

EN: The stock has delta 1. A forward struck at K is worth $Se^{(-q\tau)} - Ke^{(-r\tau)}$, so its delta is $e^{(-q\tau)}$, which is 1 without dividends. A future is settled daily, so its delta to spot is $e^{((r-q)\tau)}$, slightly above 1; hedging with futures therefore needs a tailing adjustment.

中文: 股票本身是 1。行权价为 K 的 forward 价值是 $Se^{(-q\tau)} - Ke^{(-r\tau)}$ ，delta 是 $e^{(-q\tau)}$ ，无分红时为 1。Futures 每日结算，对即期价格的 delta 是 $e^{((r-q)\tau)}$ ，略大于 1，所以用 futures 对冲要做 **tailing** 调整。

Q46. You're long 100 calls with delta 0.6 (each on 100 shares). How do you delta hedge?

EN: Position delta = $100 \times 100 \times 0.6 = 6,000$ shares, so sell 6,000 shares. As the stock rises, delta grows and you sell more; as it falls, you buy back. Hedging a long gamma position means buying low and selling high, which is where gamma scalping profits come from (Q78).

中文：头寸 $\Delta = 100 \times 100 \times 0.6 = 6,000$ 股，所以卖出 6,000 股。标的上涨、 Δ 变大就继续卖；标的下跌就买回。Long gamma 头寸的对冲就是**低买高卖**，这是 gamma scalping 的来源（Q78）。

Q47. What's the difference between sticky-strike and sticky-delta, and why does it matter for delta?

EN: BS delta assumes vol is unchanged when spot moves. With skew, a spot move also moves the vol you should use. Sticky strike: each strike keeps its implied vol. Sticky delta (sticky moneyness): the smile moves with spot. The two give different effective deltas. On equity indices with negative skew, vol rises as the index falls, so a long call's effective delta is usually below the BS delta; this is the smile-adjusted (shadow) delta.

$$\Delta_{\text{eff}} = \Delta_{\text{BS}} + \text{Vega} \cdot \frac{\partial \sigma_{\text{imp}}}{\partial S}$$

中文：BS delta 默认标的变动时波动率不变，但有 skew 时标的变动会带动 vol 一起变。**Sticky strike**：每个行权价的 implied vol 不变。**Sticky delta**：vol 跟着 moneyness 走，整条曲线随标的的平移。两种假设得出的有效 delta 不同。股指是负 skew，指数下跌时 vol 上升，所以 long call 的有效 delta 通常小于 BS delta，对应术语是 **smile-adjusted delta** 或 **shadow delta**。

Q48. Is a delta-neutral portfolio risk-free?

EN: No. Delta neutrality only hedges small first-order moves. You still carry gamma (P&L from large moves), vega (vol changes), theta (time decay), and jump, correlation and liquidity risk. And delta keeps changing, so without rebalancing you stop being neutral.

中文：不是。Delta neutral 只对冲了小幅价格变动的一阶风险，仍然暴露于：**Gamma**（大幅变动的二阶损益）、**Vega**（波动率变化）、**Theta**（时间流逝），以及跳跃、相关性和流动性风险。而且 delta 一直在变，不再平衡就不再中性。

6. Gamma & Theta (Q49–Q55)

Q49. What is gamma and why is it highest for ATM options near expiry?

EN: Gamma is the sensitivity of delta to the underlying, i.e. the convexity of the price curve. Just before expiry, an ATM option's delta can jump from 0 to 1 over a tiny price move, so gamma is huge. In the formula $\Gamma = \phi(d1)/(S\sigma\sqrt{\tau})$, $\phi(d1)$ peaks at the money and the denominator goes to 0 as $\tau \rightarrow 0$. Away from the money $\phi(d1)$ decays faster than the denominator, so gamma goes to 0. Follow-up: as expiry nears, ATM gamma rises but OTM gamma falls.

中文：Gamma 是 delta 对标的的敏感度，衡量价格曲线的**凸性**。到期前 ATM 期权的 delta 可能在极小的价格变化内从 0 跳到 1，所以 gamma 极大。公式上，ATM 时 $\phi(d1)$ 最大， $\tau \rightarrow 0$ 时分母趋近 0；远离 ATM 时 $\phi(d1)$ 衰减得比分母快，gamma 趋近 0。常见追问：随着到期临近，

ATM gamma 变大，OTM gamma 反而变小。

Q50. Can gamma be negative for a long vanilla option?

EN: No. A vanilla payoff is convex, and its density-weighted average stays convex, so long calls and long puts both have positive, equal gamma. Gamma can be negative for short options, one side of a digital, a barrier near its barrier, and the high-strike side of a call spread.

中文：不能。Vanilla payoff 是凸函数，按密度加权后仍然凸，所以 long call 和 long put 的 gamma 都为正且相等。Gamma 可以为负的情形：short 期权、digital 的一侧、障碍期权靠近障碍时、call spread 的高行权价一侧。

Q51. What is theta? Is it always negative?

EN: Theta is the change in option value as time passes, all else equal. It's usually negative for long vanillas because time value is bleeding away. Exceptions: a deep ITM European put can have positive theta, because it behaves like $Ke^{-r\tau} - S$ and gains as discounting shrinks (the same reason American puts get exercised early); a deep ITM European call on a high-dividend stock can too.

中文：Theta 是期权价值随时间流逝的变化，其他条件不变。Long vanilla 通常为负，因为时间价值在流失。例外：**深度 ITM 的欧式 put** theta 可以为正，它的价值近似 $Ke^{-r\tau} - S$ ，随时间推移折现减少、价值上升（也解释了美式 put 为何会提前行权）；**高分红股的深度 ITM 欧式 call** 同样可能为正。

Q52. Explain the gamma-theta relationship.

EN: Plugging a delta-hedged portfolio into the BS PDE gives the relation below; ignoring rates, $\Theta \approx -\frac{1}{2}\sigma^2 S^2 \Gamma$. Long gamma always pays theta and short gamma always collects it: there is no free convexity. Theta is the rent you pay for gamma, and long gamma wins if realised vol exceeds implied.

$$\Theta + \frac{1}{2} \sigma^2 S^2 \Gamma = r(V - \Delta S)$$

中文：把 delta-hedged 组合代入 BS PDE 得到上式，忽略利率后就是 $\Theta \approx -\frac{1}{2}\sigma^2 S^2 \Gamma$ 。Long gamma 必然付 theta，short gamma 必然收 theta，也就是“**没有免费的凸性**”。Theta 是你为 gamma 付的租金；实际波动率高于隐含波动率时，gamma 赚到的超过租金。

Q53. You are short gamma. What keeps you up at night?

EN: Large, sudden moves. In a gap you can't hedge in time, and each rebalance means buying high and selling low. Near expiry with spot at the strike there's also pin risk. You collect a little theta every day and can lose a lot in one jump: picking up pennies in front of a steamroller. Mitigants: gamma limits, buying OTM wings for tail protection, and stress testing.

中文：大幅、突然的价格变动。大跳空时对冲跟不上，每次再平衡都是高买低卖。临近到期、标的靠近行权价时还有 **pin risk**。每天收到的 theta 很少，一次跳跃的损失却可能很大，典型的“在压路机前捡硬币”。应对：设 gamma 限额、买入 OTM wings 对冲尾部、做压力测试。

Q54. How do you hedge gamma? Can you hedge it with the stock?

EN: Not with the stock, whose gamma is zero; you need other options. First use options to bring gamma (and often vega) to target, which changes your delta; then re-hedge delta with the stock. To hedge gamma and vega together you need options of at least two maturities, because gamma concentrates in short maturities and vega in long ones.

中文：不能用标的对冲，因为标的的 gamma 为 0，必须用其他期权。步骤：先用期权把 gamma（有时连同 vega）调到目标水平，这会改变组合 delta；再用标的把 delta 调回 0。要同时对冲 gamma 和 vega，至少需要两个不同到期日的期权，因为 gamma 集中在短期限，vega 集中在长期限。

Q55. Compare gamma and vega across maturities.

EN: For ATM options, $\Gamma \propto 1/\sqrt{\tau}$ and $\text{Vega} \propto \sqrt{\tau}$. Short-dated options are high gamma, low vega: a realised-vol trade. Long-dated options are the opposite: an implied-vol trade. A calendar spread (short near, long far) is short gamma and long vega, a bet on a quiet market with rising implied vol.

中文：ATM 时 $\Gamma \propto 1/\sqrt{\tau}$ 、 $\text{Vega} \propto \sqrt{\tau}$ 。短期限 gamma 大、vega 小，是实际波动率的交易；长期限正好相反，是隐含波动率的交易。Calendar spread（short 近月、long 远月）是 short gamma、long vega，赌近期标的平静、同时隐含波动率上升。

7. Vega & 波动率 / Volatility (Q56–Q63)

Q56. What is vega? Where is it largest?

EN: Vega is the sensitivity of the option price to implied vol, $S\phi(d1)\sqrt{\tau}$, the same for calls and puts by parity. Across strikes it peaks near the money (where $d1 = 0$); across maturities it grows with $\sqrt{\tau}$. Trivia: vega isn't a Greek letter.

中文：Vega 是期权价格对隐含波动率的敏感度，公式 $S\phi(d1)\sqrt{\tau}$ ，由平价关系 call 和 put 相同。按行权价看在 ATM（ $d1 = 0$ 处）最大；按期限看随 $\sqrt{\tau}$ 增长。小常识：Vega 不是真正的希腊字母。

Q57. What is implied vs realised (historical) volatility?

EN: Implied vol is backed out from market option prices; it's forward-looking and includes a risk premium. Realised vol is the annualised standard deviation of log returns; it's backward-looking. Implied usually exceeds subsequently realised vol, and that gap is the volatility risk premium: selling options earns it on average, at the cost of tail risk.

$$\sigma_{\text{realised}} = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (r_i - \bar{r})^2}, \quad r_i = \ln \frac{S_i}{S_{i-1}}$$

中文：Implied vol 是从市场期权价格反解 BS 得到的，向前看，包含风险溢价。Realised vol 是对数收益率的年化标准差，向后看。隐含通常高于之后实现的波动率，差额叫 volatility risk premium。卖期权长期有正收益，但尾部风险很大。

Q58. Why is there a volatility smile/skew? Explain equity skew specifically.

EN: BS assumes one vol for all strikes; markets don't. Equity indices show negative skew (low-strike puts are dearer) because of (1) the leverage effect: falling prices raise leverage and vol; (2) crash fear since 1987: returns are negatively skewed and fat-tailed; (3) supply and demand: institutions keep buying puts and selling covered calls. FX smiles are more symmetric since either currency can move sharply; commodities, especially agricultural, often show positive skew because supply shocks cause spikes.

中文：BS 假设所有行权价波动率相同，现实并非如此。股指呈负 skew（低行权价 put 更贵），原因是：① **Leverage effect**：股价下跌使杠杆率升高，波动率随之上升；② **崩盘恐惧**：1987 年后市场对左尾定价，收益分布负偏、厚尾；③ **供需**：机构持续买 put 保护、卖 covered call。FX 更接近对称的 smile；商品（尤其农产品）常见正 skew，因为供给冲击会导致价格暴涨。

Q59. What do risk reversal and butterfly measure?

EN: In FX quoting, the 25Δ risk reversal = σ(25Δ call) – σ(25Δ put) measures skew, i.e. which side the market favours. The 25Δ butterfly = ½[σ(25Δ call) + σ(25Δ put)] – σ(ATM) measures the smile's curvature, i.e. how fat the tails are. ATM, RR and BF together pin down the smile for one maturity.

中文：这是 FX 报价体系。**25Δ risk reversal** = σ(25Δ call) – σ(25Δ put)，衡量 skew（市场偏向哪一侧）。**25Δ butterfly** = ½[σ(25Δ call) + σ(25Δ put)] – σ(ATM)，衡量 smile 的弯曲度，也就是尾部有多厚。有了 ATM、RR 和 BF 这三个数，就能构造一个期限上的 smile。

Q60. What is the volatility term structure and why does it usually slope upward?

EN: In calm markets short-dated vol sits below the long-run average, so the term structure slopes up (contango), driven by vol mean reversion plus an uncertainty premium at longer maturities. In a crisis short-dated vol spikes and the curve inverts (backwardation). The VIX futures curve shows this clearly.

中文：平静市场中，短期波动率低于长期均值，期限结构向上倾斜（contango），原因是波动率均值回归，加上长期限的不确定性溢价。危机时短期波动率飙升，期限结构倒挂（backwardation）。VIX 期货曲线就是这个现象的直观体现。

Q61. How do you compute implied volatility? Why can't you invert BS analytically?

EN: The BS price is transcendental in σ , so there's no closed-form inverse; solve numerically. Newton-Raphson converges fast, but vega $\rightarrow 0$ for deep OTM options, so iterations can diverge. Robust practice: fall back to Brent or bisection; for speed and precision use Jäckel's "Let's be rational". Check first that the price is inside the no-arbitrage bounds (Q27).

$$\sigma_{n+1} = \sigma_n - \frac{C_{BS}(\sigma_n) - C_{mkt}}{\text{Vega}(\sigma_n)}$$

中文：BS 价格关于 σ 是超越函数，没有闭式反函数，只能数值求解。Newton-Raphson 收敛很快，但深度 OTM 时 vega 趋近 0，迭代会发散。稳健做法是用 Brent 或二分法兜底；精确且快的做法是 Jäckel 的“Let's be rational”。迭代前先检查价格是否在 Q27 的无套利区间内。

Follow-up / 追问： Why does Newton-Raphson sometimes fail for implied vol, and how do you fix it?

EN: Newton's step is (price error) $\div$ vega, so it breaks when vega is tiny or the starting point is on the wrong side of the curve. Example: $S = 100$, $K = 130$, $T = 0.1$, true vol 40%, option price ≈ 0.10 . Starting at 20%, the deep OTM option has almost no vega, so the first step jumps to about 3,800% vol, the next goes negative, and the iteration ends in NaN. Starting at the inflection point below instead converges to 40.00% in about six steps, and Brent's method on [0.01%, 500%] finds it without a derivative.

中文：Newton 的每一步是“价格误差 $\div$ vega”，所以 vega 很小、或者起点在曲线“错的一侧”时就会出问题。例子： $S = 100$ 、 $K = 130$ 、 $T = 0.1$ 、真实 vol 40%，期权价格约 0.10。从 20% 开始迭代，深度 OTM 期权几乎没有 vega，第一步就跳到约 3,800% 的波动率，下一步变成负数，最后得到 NaN。改从下面的拐点开始，大约 6 步收敛到 40.00%；用 Brent 法在 [0.01%, 500%] 区间求解也能找到，且不需要导数。

Problem / 问题	Why it happens / 原因	Fix / 解决办法
Vega ≈ 0 / vega 接近 0	Deep ITM/OTM or very short expiry: price barely reacts to vol / 深度 ITM/OTM 或极短期限，价格对 vol 几乎不敏感	Bracket and fall back to bisection or Brent / 加区间，失败时退回二分或 Brent
Bad starting point / 起点不好	Price is convex in σ below an inflection point and concave above it, so Newton can overshoot / 价格关于 σ 在拐点下方凸、上方凹，Newton 会冲过头	Start at the inflection point (Manaster-Koehler) or a Brenner-Subrahmanyam / Corrado-Miller guess / 从拐点或近似公式出发

Problem / 问题	Why it happens / 原因	Fix / 解决办法
Step leaves valid range / 步长越界	Negative or absurd σ after one step / 一步后出现负数或离谱的 σ	Safeguarded Newton: keep a bracket, reject steps outside it (like rtsafe) / 带保护的 Newton
No solution / 无解	Price outside no-arbitrage bounds (Q27) / 价格超出无套利区间	Check bounds first and flag the quote / 先检查边界，标记异常报价
Precision loss near intrinsic / 接近内在价值时精度损失	Deep ITM price = intrinsic + tiny time value; the difference cancels / 深度 ITM 时间价值极小，相减丢精度	Invert the OTM option via parity; use erfc for tail probabilities / 用 parity 换成 OTM 期权反解；尾部概率用 erfc
Slow or fragile convergence / 收敛慢或不稳	Price is very nonlinear in σ for far strikes / 远行权价时价格关于 σ 高度非线性	Iterate on $\ln(\text{price})$ or total vol $\sigma\sqrt{T}$; use Halley's method (adds volga); or Jäckel's "Let's be rational" / 对 $\ln(\text{价格})$ 或 $\sigma\sqrt{T}$ 迭代，用 Halley 法或 Jäckel 算法

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\sigma^{\ast} = \sqrt{\frac{2\,|\ln(F/K)|}{T}} \quad \text{(inflection point: Newton from here converges monotonically)}
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EN: Why the inflection point works: at σ^* , vega is at its maximum along the vol axis, so the price curve is steepest there. On either side of it the curve bends in a single direction, so a Newton step from σ^* never overshoots, and the iterates move steadily towards the answer. A production solver typically does four things: check the no-arbitrage bounds, switch to the OTM option, start from a good analytic guess, then run safeguarded Newton or Halley with a bisection fallback, a price and vol tolerance and a maximum iteration count, and flag failures for review rather than returning a number. Jäckel's algorithm (2015) reaches machine precision in two iterations and is the industry reference.

中文：为什么从拐点出发有效：在 σ^* 处，沿波动率方向的 vega 最大，价格曲线最陡；拐点两侧曲线各自只往一个方向弯，所以从 σ^* 出发的 Newton 步不会冲过头，迭代值会单调地靠近答案。生产系统里的求解器一般做四件事：① 检查无套利边界；② 换成 OTM 期权；③ 从较好的解析近似出发；④ 用带保护的 Newton 或 Halley 迭代，配二分兜底、价格和波动率的容差、最大迭代次数，失败时标记待查，而不是硬返回一个数。Jäckel 的算法（2015）两次迭代就达到机器精度，是业内标杆。

Q62. Vega of a long straddle vs a calendar spread vs a butterfly?

Structure / 结构	Vega	Note / 说明
Long straddle	Strongly + / 大幅为正	Two ATM vegas add up / 两个 ATM vega 相加
Calendar (short near, long far)	+	Far-month vega is larger / 远月 vega 更大
Long butterfly	- near the body / ATM 附近为负	Long wings, short 2 × body; volga +
Risk reversal	≈ 0	Exposed to skew (vanna) / 暴露在 skew 上

EN: The logic in each case is simply summing the vegas of the legs, remembering vega is largest ATM and grows with maturity.

中文：每种结构的逻辑都是把各腿的 vega 相加，记住 vega 在 ATM 最大、随期限增长即可。

Q63. How does the vol surface feed into risk management?

EN: Each position maps to a (strike, maturity) point on the surface. Beyond total vega, decompose into vega by maturity bucket, skew and smile sensitivities (RR and BF vega), and parallel, tilt and curvature moves of the surface. On the valuation side, compare the desk's surface with independent sources (Totem, broker quotes) as IPV, and hold model or bid-offer reserves for illiquid regions. See section 12.

中文：每个头寸对应 surface 上一个（行权价，期限）点。风险分解不只看总 vega，还要看：**Vega by bucket**（按期限分桶）、**Skew 和 smile 敏感度**（RR、BF 的 vega）、surface 的平行移动、斜率和弯曲变化。Valuation 方面要把交易员的 surface 和独立来源（Totem、broker quote）比对（IPV），不流动区域计提 model reserve 或 bid-offer reserve。这是你 valuation 背景的分题，见第 12 节。

FX 波动率曲面深入 / FX volatility surface deep dive (FX1-FX7)

EN: This extends Q59. FX vol is quoted by tenor × delta, and each tenor's smile is described by three numbers: ATM (the level), the risk reversal (the tilt) and the butterfly (how curved the wings are).

中文：这是 Q59 的展开。FX 波动率曲面按**期限 × delta** 报价，每个期限用三个数描述一条 smile：ATM（水平）、risk reversal（倾斜方向）、butterfly（两翼弯曲程度）。

FX1. Why is the FX vol surface quoted by delta instead of strike?

EN: The axes are tenor (O/N, 1W, 1M ... 1Y, 2Y) and delta (10Δ put, 25Δ put, ATM, 25Δ call, 10Δ call). Delta already contains $\sigma\sqrt{T}$, so the same delta means the same distance from the forward in standard deviations across tenors and pairs ($25\Delta \approx 0.67 \text{ sd}$). It fits sticky-delta

behaviour, where the smile moves with spot. And the quoted points are tradable structures: ATM straddles, 25Δ risk reversals and 25Δ strangles trade every day.

中文：纵轴是期限，横轴是 delta。用 delta 而不用行权价，有三个原因：① **可以跨期限、跨货币对比较**：delta 已经包含 $\sigma\sqrt{T}$ ，同样是 25Δ，1 周的行权价离 forward 很近、1 年的远得多，但“离 forward 几个标准差”相同；② **符合 sticky-delta 的市场行为**：即期汇率变动后 smile 大体跟着 moneyness 平移；③ **报出的都是可交易的结构**。

FX2. What quoting conventions do you need to know?

EN: (1) Spot vs forward delta: G10 pairs usually use spot delta up to 1 year and forward delta beyond; many EM pairs use forward delta throughout. (2) Premium-adjusted delta depends on the premium currency: EURUSD premium is paid in USD (the quote currency), so delta is unadjusted; USDJPY premium is paid in USD (the base currency), so delta includes the premium. (3) ATM definition: most pairs use ATM DNS (delta-neutral straddle); some EM pairs and long tenors use ATMF (strike = forward).

中文：① **Spot delta 与 forward delta**：G10 在 1 年以内通常用 spot delta，更长期限用 forward delta；很多 EM 货币对一律用 forward delta。② **是否包含期权费**（premium-adjusted delta）取决于期权费用哪种货币支付：EURUSD 用 USD（报价货币）支付，delta 不调整；USDJPY 用 USD（基础货币）支付，delta 要包含期权费。③ **ATM 的定义**：大多数货币对用 ATM DNS，少数 EM 或长期限用 ATMF。

FX3. What is ATM in FX, and how do the ATM call and ATM put differ?

$$K_{\text{DNS}} = F \cdot e^{+\frac{1}{2}\sigma^2 T}; \text{unadjusted}, \quad K_{\text{DNS}} = F \cdot e^{-\frac{1}{2}\sigma^2 T}; \text{premium-adjusted}$$

EN: ATM DNS is the strike where the call and put deltas are equal and opposite, so the straddle is delta-neutral; it sits very slightly above the forward (unadjusted) or below it (premium-adjusted). By put-call parity the call and put at the same strike share one implied vol, so ATM is a single number; their prices differ only by $PV(F - K)$, which is tiny at the DNS strike. In FX a EUR call is the same contract as a USD put, so always say whose call. ATM is the level of vol (σ in a model) and is traded with straddles: pure vega and gamma, delta about zero.

中文：ATM DNS 是让 call 和 put 的 delta 正好相反的行权价，所以 straddle 的 delta 为 0；它略高于 forward（不调整）或略低于 forward（包含期权费）。同一行权价上，由 put-call parity，call 和 put 的 implied vol 必须相同，所以 ATM vol 只有一个数；价格差等于 $PV(F - K)$ ，在 DNS 行权价上几乎为 0。FX 的特点是 **EUR call 就是 USD put**，说“call 更贵”时一定要说明是哪种货币的 call。ATM 是波动率水平（模型里的 σ ），用 straddle 交易，是纯 vega 加 gamma 头寸。

FX4. What does the risk reversal measure, and how is it traded?

$$\text{RR}_{25} = \sigma_{25\Delta, \text{call}} - \sigma_{25\Delta, \text{put}}$$

EN: The RR measures which way the smile tilts, i.e. which direction of a big move the market pays more to protect. $RR > 0$: calls on the base currency are richer; $RR < 0$: puts are richer. Typical logic (signs move with markets): USD/EM pairs often have positive RR because the market fears sudden EM depreciation; USDJPY often has negative RR because the yen rallies in a panic, so JPY calls (USD puts) are bid. A long RR is long the 25Δ call and short the 25Δ put: vega roughly zero, delta positive, main exposure vanna. In models it maps to spot-vol correlation (ρ in SABR or Heston).

中文：RR 衡量 smile 向哪边倾斜，即市场更愿意为哪个方向的大幅波动付钱。 $RR > 0$ 表示基础货币的 call 更贵； $RR < 0$ 表示 put 更贵。典型含义（方向随行情变化，这里只讲逻辑）：USD/EM 货币对的 RR 常为正，因为市场怕新兴市场货币突然贬值；USDJPY 常见 RR 为负，因为日元是避险货币，恐慌时急涨，JPY call（USD put）更贵。交易结构是 long 25Δ call + short 25Δ put：vega 约为 0、delta 为正、主要暴露是 **vanna**。模型里对应即期汇率与波动率的相关系数 ρ 。

FX5. What does the butterfly measure, and what is the market-strangle subtlety?

$$\text{BF}_{25} \approx \frac{1}{2} \left(\sigma_{25\Delta, \text{call}} + \sigma_{25\Delta, \text{put}} \right) - \sigma_{\text{ATM}}$$

EN: The BF measures how expensive the wings are relative to ATM, i.e. how curved the smile is and how fat the tails are. It maps to kurtosis in the distribution and to vol of vol in models (v in SABR). The trade is a long 25Δ strangle against a short ATM straddle, sized to be vega-neutral; the main exposure is volga. Advanced point: the broker BF is a market-strangle quote. A single vol, $\sigma_{\text{ATM}} + \text{BF}_{\text{mkt}}$, prices both the 25Δ call and put, and the strikes are solved with that same vol. The true smile has different vols for the two wings (they differ by the RR), so the smile must be built to reprice that market strangle. The simple formula above is only an approximation, and the gap grows with the RR.

中文：BF 衡量两翼相对 ATM 有多贵，即 smile 的弯曲程度、尾部有多厚；在分布上对应峰度，在模型里对应 vol of vol（SABR 的 v ）。交易结构是 long 25Δ strangle、short ATM straddle，并调整数量使总 vega 为 0，主要暴露是 **volga**。**高阶点：**经纪商报的 BF 是 **market strangle** 的报价，用一个统一的波动率 $\sigma_{\text{ATM}} + \text{BF}_{\text{mkt}}$ 同时给 25Δ call 和 put 定价，行权价也用这个波动率反解。真实 smile 上两翼的波动率不同（相差 RR），构建 smile 时必须让它给出的 strangle 价格等于这个市场报价，所以上面的公式只是近似，RR 越大差得越多。

FX6. How do you rebuild the smile from ATM, RR and BF?

$$\begin{aligned} \sigma_{25\Delta, \text{call}} &= \sigma_{\text{ATM}} + \text{BF} + \frac{1}{2} \text{RR}, \\ \sigma_{25\Delta, \text{put}} &= \sigma_{\text{ATM}} + \text{BF} - \frac{1}{2} \text{RR} \end{aligned}$$

EN: Example, EURUSD 1M: ATM 8.0%, 25 Δ RR -1.0%, 25 Δ BF 0.3% give a 25 Δ call vol of 7.8% and a 25 Δ put vol of 8.8%. Converting deltas to strikes with $F = 1.1000$ and forward delta gives the table below. Do the same for 10 Δ to get five points per tenor, then interpolate within a tenor with Vanna-Volga, SABR or a polynomial in delta. Across tenors interpolate total variance $\sigma^2 T$, weighting time for events: weekends and holidays count less, central-bank meetings more.

Point / 点	Vol / 波动率	Strike / 行权价
25 Δ put	8.8%	≈ 1.0817
ATM DNS	8.0%	≈ 1.1003
25 Δ call	7.8%	≈ 1.1171

中文：例子：EURUSD 1M，ATM 8.0%、25 Δ RR -1.0%、25 Δ BF 0.3%，得到 25 Δ call 7.8%、25 Δ put 8.8%。再由 delta 反解行权价（ $F = 1.1000$ ，按 forward delta）得到上表。同样还原 10 Δ ，每个期限就有 5 个点，再用 Vanna-Volga、SABR 或按 delta 的多项式插值成完整的 smile。期限之间对总方差 $\sigma^2 T$ 插值，并按事件调整时间权重：周末、节假日权重低，央行议息等事件日权重高。

FX7. Compare ATM, RR and BF. How are they used in practice?

	ATM	Risk reversal	Butterfly
Measures / 衡量	Vol level / 波动率水平	Tilt (skew) / 倾斜方向	Wing curvature, tail thickness / 两翼弯曲、尾部厚度
Distribution / 分布含义	Standard deviation / 标准差	Skewness / 偏度	Kurtosis / 峰度
Model analogue / 模型对应	σ	ρ (spot-vol correlation)	v (vol of vol)
Trade / 交易结构	Straddle	Long call + short put	Strangle vs vega-neutral straddle
Main Greek / 主要 Greek	Vega, gamma	Vanna (vega ≈ 0)	Volga
Sign / 符号	Always positive / 总是正	+ calls richer, - puts richer / 正：call 贵，负：put 贵	Usually positive; larger = fatter tails / 通常为正

EN: In practice: (1) Vanna-Volga pricing uses exactly these three instruments to hedge and price FX exotics such as barriers and digitals. (2) FX option books report ATM vega, RR (vanna) and BF (volga) by tenor rather than one total vega. (3) IPV (Q93) compares ATM, 25Δ/10Δ RR and BF per tenor with Totem or broker quotes; G10 tolerances are often about 0.25–0.5 vol points each. (4) FX smiles are more symmetric than equity index smiles because either currency can jump, so BF is significant and the direction of skew depends on the RR sign; equity indices mainly fear falls, giving one-sided negative skew.

中文：实务用途：① **给奇异期权定价**：Vanna-Volga 方法正是用 ATM、RR、BF 这三种结构来对冲并定价 FX 奇异期权；② **风险管理**：按期限汇报 ATM vega、RR (vanna) 和 BF (volga)，而不是只看一个总 vega；③ **IPV (Q93)**：按期限把 ATM、25Δ/10Δ RR 和 BF 与 Totem 或经纪商报价比对，G10 常见容差各约 0.25–0.5 个 vol 点；④ **为什么 FX 的 smile 比股指更对称**：两种货币都可能大幅升值或贬值，所以两边都翘起 (BF 明显)，skew 方向看 RR 的符号；股指主要担心下跌，是单边的负 skew。

波动率风险溢价 / Volatility risk premium (VRP1-VRP6)

EN: This extends Q57. Implied vol is, on average, higher than the vol that is later realised. The gap is the volatility risk premium: option sellers are paid for insuring others against large moves, and they pay out heavily when those moves arrive.

中文：这是 Q57 的展开。平均来看，隐含波动率高于之后实际实现的波动率，这个差额就是**波动率风险溢价 (VRP)**：卖期权的人就像保险公司，平时收保费，大行情来了就要大额赔付。

VRP1. What is the volatility risk premium, and how large is it?

$$\text{VRP}_t = \sigma^{\{\text{implied}\}}_{t,T} - \sigma^{\{\text{realised}\}}_{t,T}, \quad \text{variance version: } \mathbb{E}^{\{\mathbb{Q}\}}[RV_{t,T}] - \mathbb{E}^{\{\mathbb{P}\}}[RV_{t,T}]$$

EN: It's the difference between the vol the market charges today and the vol that actually materialises over the option's life; academics often define it in variance terms, as the gap between risk-neutral and real-world expected variance (Carr & Wu, 2009). For the S&P 500, 30-day implied vol (VIX) has historically averaged roughly 3–4 vol points above subsequent realised vol, and implied has exceeded realised in most months, roughly four in five. These figures are approximate long-run averages; the premium varies a lot over time and across markets, and it's usually larger for index options than single stocks and for OTM puts than ATM options.

中文：VRP 是市场今天收取的波动率与期权存续期内实际发生的波动率之差；学术上常用方差定义，即风险中性期望方差与真实世界期望方差之差 (Carr & Wu, 2009)。以标普 500 为例，30 天隐含波动率 (VIX) 历史上平均比之后的实现波动率高约 3–4 个 vol 点，大约五个月里有四个月隐含高于实现。这些是大致的长期平均，溢价随时间和市场变化很大；通常指数期权比个股大，OTM put 比 ATM 大。

VRP2. Why does the premium exist, and why isn't it arbitrated away?

EN: (1) Insurance demand: pension funds, asset managers and structured-product hedgers are natural buyers of downside protection. (2) Sellers bear crash risk: short vol loses exactly when everything else loses, so it pays badly in bad states and investors demand compensation, the same logic as any insurance premium. (3) Jumps and vol-of-vol can't be hedged by delta hedging, so the risk is genuinely unhedgeable. (4) Limits to arbitrage: dealers who warehouse short vol face capital, margin and VaR constraints that bind precisely in a crisis. Flows can also push the other way in places: structured products such as autocallables leave dealers long long-dated vol, which tends to compress long-dated implied vol.

中文：① **保险需求：**养老金、资产管理人和结构化产品对冲方天然要买下行保护。② **卖方承担崩盘风险：**short vol 恰好在所有资产都亏的时候亏，在最糟糕的状态下赔付，所以要求补偿，和任何保险溢价的逻辑一样。③ **跳跃和 vol-of-vol 无法通过 delta 对冲消除，**是真正无法对冲的风险。④ **套利有限：**持有 short vol 的交易商受资本、保证金和 VaR 限制，而这些限制恰好在危机时收紧。也有反向力量：autocallable 等结构化产品让交易商持有长期限 long vol，往往压低长期隐含波动率。

VRP3. How do you trade the volatility risk premium?

Instrument / 工具	How it harvests VRP / 怎么赚溢价	Main risk / 主要风险
Delta-hedged short straddle or strangle / 做 delta 对冲的卖出跨式	Earns $\frac{1}{2}\Gamma S^2(\sigma_{\text{implied}}^2 - \sigma_{\text{realised}}^2)$ each day (Q77) / 每天赚方差差	Path-dependent, gamma near expiry, hedging costs / 路径依赖、临近到期 gamma
Short variance swap / 卖出方差互换	Pure exposure: payoff $N_{\text{var}} \times (K_{\text{var}} - \sigma_{\text{R}}^2)$	Convex losses in a crash (variance, not vol) / 崩盘时损失按方差放大
Short VIX futures / 卖出 VIX 期货	Roll-down in contango as futures converge to spot VIX / 升水结构下的展期收益	Violent spikes, term-structure inversion / 暴涨和期限结构倒挂
Put writing, covered calls (e.g. CBOE PUT, BXM)	Collects premium without hedging / 不对冲直接收期权费	Also carries equity beta / 同时承担股票 beta
Relative value: dispersion, calendar, skew / 相对价值	Sells the richest part of the surface against a cheaper part / 卖最贵的一段、买便宜的一段	Correlation spikes, basis risk / 相关性飙升、基差

$$\text{Var swap payoff (long)} = N_{\{\text{var}\}} \cdot (\sigma_R^2 - K_{\{\text{var}\}}), \quad N_{\{\text{vega}\}} = N_{\{\text{var}\}} \cdot 2K_{\{\text{vol}\}}$$

EN: Variance-swap strikes sit above ATM vol because they are replicated with a $1/K^2$ -weighted strip of options, which picks up the expensive downside skew. That's one reason short variance often earns more than short ATM straddles, and also why it loses more in a crash. Practitioners size by stress loss rather than volatility, keep long wings (put spreads, iron condors instead of naked straddles), and buy some tail protection (far OTM puts or VIX calls) with part of the premium.

中文：方差互换的行权价高于 ATM vol，因为它用按 $1/K^2$ 加权的一组期权复制，包含了昂贵的下行 skew。这是 short variance 往往比 short ATM straddle 赚得多的原因，也是它崩盘时亏得更多的原因。实务中按**压力测试损失**而不是按波动率定仓位；保留两翼多头（用 put spread、iron condor 替代裸卖 straddle）；并用一部分溢价买尾部保护（深度 OTM put 或 VIX call）。

VRP4. When does selling volatility fail?

EN: Short vol has negatively skewed returns: many small gains, then rare large losses, the “pennies in front of a steamroller” profile. It fails when: (1) realised vol jumps above implied, typically when vol rises from a low base or on a sudden shock; (2) the premium has been compressed by crowded selling, leaving little cushion; (3) event or jump risk is underpriced; (4) the term structure inverts and short VIX futures lose on both spot and roll; (5) liquidity dries up and margin calls force short-vol holders to cover at the worst time, which pushes vol higher still; (6) correlations go to 1, which breaks dispersion trades. Historical episodes include October 1987, late 2008 (VIX closed above 80), the February 2018 “Volmageddon” when VIX more than doubled in a day and the inverse-VIX note XIV collapsed and was terminated, March 2020 (VIX closed at a record of about 83), and August 2024, when the yen carry-trade unwind briefly pushed intraday VIX to around 65.

中文：short vol 的收益分布是负偏的：平时赚很多小钱，偶尔亏一笔大钱，典型的“在压路机前捡硬币”。失效的情形：① **实现波动率跳到隐含之上**，常见于波动率从低位抬升或突发冲击；② **溢价被拥挤的卖方压缩**，安全垫太薄；③ **事件或跳跃风险定价不足**；④ **期限结构倒挂**，short VIX 期货在现货和展期上双重亏损；⑤ **流动性枯竭、保证金追缴**，迫使 short vol 持有者在最差的时点平仓，反过来进一步推高波动率；⑥ **相关性趋向 1**，dispersion 交易失效。历史案例：1987 年 10 月；2008 年末（VIX 收盘超过 80）；2018 年 2 月的 “Volmageddon”，VIX 一天内翻倍以上，反向 VIX 产品 XIV 暴跌后被终止；2020 年 3 月（VIX 收盘创约 83 的纪录）；2024 年 8 月日元套息交易平仓，VIX 盘中一度冲到 65 左右。

VRP5. How do you judge whether the premium is attractive right now?

EN: Compare implied vol with a forecast of realised vol, not with past realised alone (e.g. GARCH or HAR-type forecasts, or recent realised with a mean-reversion adjustment). Check the VIX term structure: steep contango is supportive, flat or inverted is a warning. Watch vol-of-vol (VVIX), skew and positioning, since crowded short vol means a thin cushion. Judge performance on risk-adjusted terms that respect the tails, such as drawdown and Expected Shortfall, because Sharpe ratios of short-vol strategies look great until the crash.

中文：用隐含波动率和对未来实现波动率的预测比，而不是只和过去的实现波动率比（比如 GARCH、HAR 类预测，或者近期实现波动率加均值回归调整）。看 VIX 期限结构：陡峭的升水是有利信号，平坦或倒挂是警告。关注 vol-of-vol (VVIX)、skew 和持仓拥挤度。评价表现要用考虑尾部的指标，比如最大回撤和 Expected Shortfall，因为 short vol 策略的 Sharpe 在崩盘前总是很好看。

VRP6. How do you risk-manage a short-volatility book?

EN: (1) Size to a stress scenario (e.g. spot -20% with vol +30 points) rather than to normal-day vol. (2) Cap vega and gamma, especially short-dated gamma into events and weekends. (3) Define risk: sell spreads rather than naked options, or pair short ATM with long OTM wings. (4) Keep a standing tail hedge and accept its cost as part of the strategy. (5) Remember diversification across underlyings largely disappears in a crisis when correlations spike. (6) Plan liquidity and margin so you are never forced to cover at the peak.

中文：① 按压力情景定仓位（比如标的 -20%、波动率 +30 个点），而不是按正常日子的波动率。② 限制 vega 和 gamma，尤其是事件和周末前的短期 gamma。③ 限定最大损失：卖价差组合而不是裸卖期权，或者在卖 ATM 的同时买 OTM 两翼。④ 常备尾部对冲，把它的成本当作策略的一部分。⑤ 记住危机时相关性飙升，跨标的的分散化基本失效。⑥ 规划流动性和保证金，保证永远不会被迫在波动率顶部平仓。

8. Rho、分红与高阶 Greeks / Rho, dividends and higher-order Greeks (Q64-Q69)

Q64. What is rho and why does a call have positive rho?

EN: Rho is sensitivity to interest rates. Higher rates raise the forward ($F = Se^{(r\tau)}$) and lower the present value of the strike; both help a call, so call rho is positive and put rho negative. Put differently, buying a call defers paying K, which is worth more when rates are high. Equity option rho is usually small, but it matters for long-dated deep ITM options and rates options; FX options have two rhos, one per currency.

中文：Rho 是期权价格对利率的敏感度。利率上升会抬高 forward，并降低行权价的现值，两个效果都对 call 有利，所以 call 的 rho 为正、put 为负。也可以理解为买 call 相当于推迟支付 K，利率越高越划算。股票期权的 rho 通常很小，但对长长期限、深度 ITM 期权和利率期权很重

要；FX 期权有两个 rho。

Q65. How do dividends affect option prices and Greeks?

EN: Dividends lower the forward ($F = Se^{((r-q)\tau)}$), so calls get cheaper and puts dearer, and call delta becomes $e^{-q\tau}N(d1)$. Discrete dividends make the stock drop on the ex-date, so American calls may be exercised just before it (Q29). Uncertainty in dividend forecasts is itself a risk, especially for long-dated index options and structured products.

中文：分红降低 forward，所以 call 变便宜、put 变贵，call 的 delta 变成 $e^{-q\tau}N(d1)$ 。离散分红会在除息日造成股价跳空，所以美式 call 可能在除息前提前行权（Q29）。分红预测的不确定性本身也是风险（dividend risk），尤其影响长期股指期权和结构化产品。

Q66. What is vanna, and why do traders care?

EN: Vanna = $\partial\Delta/\partial\sigma = \partial\text{Vega}/\partial S$: how vol changes move your delta, or how spot moves change your vega. A risk reversal is an almost pure vanna position. With negative skew, vol rises as spot falls, so a vanna position shifts delta and vega together and complicates hedging. It's also one half of Vanna-Volga pricing for FX exotics.

中文：Vanna = $\partial\Delta/\partial\sigma = \partial\text{Vega}/\partial S$ ，衡量波动率变化如何改变 delta，或标的变动如何改变 vega。Risk reversal 几乎是纯 vanna 头寸。负 skew 市场里股价下跌时波动率上升，delta 和 vega 会同时变化，对冲变复杂。它与 volga 一起是 FX exotics 的 Vanna-Volga 定价法的核心。

Q67. What is volga (vomma)?

EN: Volga = $\partial\text{Vega}/\partial\sigma$, the convexity of price in vol. ATM prices are almost linear in σ , so ATM volga is near zero; OTM volga is positive. Long volga profits from vol of vol, which is why wings trade at higher implied vol than ATM: the market charges for vol-of-vol, and that is the butterfly component of the smile.

中文：Volga = $\partial\text{Vega}/\partial\sigma$ ，即价格对波动率的凸性。ATM 价格近似关于 σ 线性，volga 接近 0；OTM 的 volga 为正。持有 long volga，波动率的波动对你有利。这解释了 wings 的 implied vol 比 ATM 高：市场要为 vol of vol 付钱，这就是 smile 的 butterfly 部分。

Q68. What are charm, speed and colour?

Greek	Definition / 定义	Practical meaning / 实务含义
Charm	$\partial\Delta/\partial t$	Delta drifts with time even if spot is flat; re-hedge after weekends / 行情不动 delta 也漂移
Speed	$\partial\Gamma/\partial S$	Gamma changes with spot; second-order approximation breaks on big moves / 大幅变动时二阶近似失效
Colour	$\partial\Gamma/\partial t$	Gamma changes with time; ATM gamma explodes near expiry / 临近到期 ATM gamma 爆炸

Greek	Definition / 定义	Practical meaning / 实务含义
Veta	$\partial \text{Vega} / \partial t$	Vega decays with time / vega 随时间衰减

EN: You don't need the formulas in an interview; explain what each means and when it matters.

中文：面试里不需要背公式，讲清楚含义和何时重要就够了。

Q69. What is the Greek of a portfolio? Are Greeks additive?

EN: For one underlying, yes: portfolio delta is the sum of position deltas. Three traps: (1) different underlyings can't be added directly; convert to money, e.g. dollar delta = $\Delta \times S$, or beta-adjusted delta; (2) adding vegas across maturities implicitly assumes a parallel term-structure shift, so desks often weight by $v(T_{\text{ref}}/T)$; (3) units must match, e.g. vega per vol point vs per 1%.

中文：对同一标的，Greeks 线性可加。三个陷阱：① 不同标的不能直接相加，要换算成货币金额（dollar delta = $\Delta \times S$ ）或 beta-adjusted delta；② **Vega 要注意期限**，不同期限直接相加隐含“整条期限结构平行移动”的假设，常按 $v(T_{\text{ref}}/T)$ 加权；③ **单位要统一**。

9. 奇异期权 / Exotics (Q70–Q75)

Q70. Price a digital (cash-or-nothing) call. How do its Greeks behave near expiry?

EN: The price is $e^{(-r\tau)}N(d_2)$ per unit paid. Its delta looks like a vanilla's gamma, peaking at the strike, and blows up as expiry nears with spot at K . Gamma changes sign: positive below the strike, negative above. Vega = $-e^{(-r\tau)}\phi(d_2)d_1/\sigma$, so its sign follows d_1 : negative once the option is in the money on a forward basis. In my CQF notebook $S = 100 < K = 105$, but $F \approx 105.1 > K$, so $d_1 > 0$ and the binary's price falls with vol (49.9 at $\sigma = 5\%$, 40.2 at $\sigma = 50\%$). Judge moneyness by the forward, not spot.

$$\Delta_{\text{digital}} = \frac{e^{-r\tau} \phi(d_2)}{S \sigma \sqrt{\tau}}$$

中文：价格是 $e^{(-r\tau)}N(d_2)$ （每单位支付）。它的 delta 形状像 vanilla 的 gamma，在行权价附近最大，临近到期且标的靠近 K 时趋向无穷。Gamma 会变号：行权价下方为正、上方为负。Vega = $-e^{(-r\tau)}\phi(d_2)d_1/\sigma$ ，符号由 d_1 决定，按 forward 看已经 ITM 时为负。你的 notebook 就是例子： $S = 100 < K = 105$ ，但 $\text{forward} \approx 105.1 > K$ ， $d_1 > 0$ ，所以 binary 价格随 σ 下降。判断 moneyness 要看 forward，不要看 spot。

Q71. How would you hedge a digital option in practice?

EN: Replicate it with a tight call spread, and shift the spread in your favour (overhedge, or bull/bear spread shift) so the payoff you hold always dominates the digital. The spread width ϵ acts as a reserve. That's far more stable than delta-hedging a product whose delta

goes to infinity before expiry.

$$\text{Digital}(K) \approx \frac{C(K-\epsilon) - C(K+\epsilon)}{2\epsilon}$$

中文：用紧密的 **call spread** 复制，并把 spread 往对自己有利的方向偏移 (overhedge)。这样 digital 的不连续性就被替换成一个总是多收钱的连续 payoff，spread 宽度 ϵ 就是风险储备。这比直接 delta 对冲一个到期前 delta 趋向无穷的产品要稳得多。

Q72. How does the volatility skew affect the price of a digital?

EN: Taking the limit of Q71, a digital call paying 1 equals $-\partial C / \partial K$. When σ depends on K , that adds a skew term. With negative equity skew ($\partial \sigma / \partial K < 0$), the digital call is more expensive than the flat-vol BS price and the digital put cheaper. This separates candidates who memorise formulas from those who understand smiles.

$$\text{Digital}_{\text{smile}} = e^{-r\tau} N(d_2) - \text{Vega}(K) \cdot \frac{\partial \sigma_{\text{imp}}}{\partial K}$$

中文：由 Q71 取极限，支付 1 的 digital call 等于 $-\partial C / \partial K$ 。当 σ 随 K 变化时会多出一项 skew 修正。股指是负 skew ($\partial \sigma / \partial K < 0$)，所以 digital call 比平波动率的 BS 价格更贵，digital put 更便宜。这是区分“会背公式”和“懂 smile”的经典题。

Q73. What is in-out parity for barriers? Describe the Greeks of an up-and-out call.

EN: Knock-in + knock-out = vanilla (same barrier, strike and expiry, no rebate). An up-and-out call is a reverse knock-out: the barrier sits in the money. Near the barrier the option is deep ITM but about to die, so delta turns sharply negative, gamma is strongly negative, and vega can be negative because higher vol raises the knock-out probability. On a touch the value jumps to zero and you must unwind a large delta hedge at once: barrier (gap) risk.

中文：Knock-in + knock-out = vanilla (同样的障碍、行权价和到期日，没有 rebate)。Up-and-out call 是 **reverse knock-out**，障碍位于 ITM 区域。靠近障碍时期权深度 ITM 却随时会归零，所以 **delta 变成大幅为负**，gamma 大幅为负，vega 可能为负（波动率越高敲出概率越大）。碰到障碍时价值跳到 0，需要一次性平掉大量 delta 对冲，这就是 **barrier risk (gap risk)**。

Q74. What is pin risk?

EN: When spot sits right at the strike at expiry, you don't know whether the option will be exercised. A short option holder can wake up on Monday long or short a full position in the underlying with no hedge, exposed to the weekend gap. Digitals and barriers have worse pin risk because their payoffs are discontinuous.

中文：到期时标的恰好停在行权价附近，就不确定期权是否会被行权。Short 期权的一方可能在周末突然多出或少掉一整份标的的头寸，周一开盘跳空时没有对冲。Digital 和障碍期权的 pin risk 更严重，因为 payoff 不连续。

Q75. Why is Monte Carlo difficult for barrier and digital Greeks, and how do you fix it?

EN: Bump-and-reprice on a discontinuous payoff differentiates an indicator: the pathwise derivative is zero almost everywhere and jumps only on paths crossing K, so the estimator is very noisy. Fixes: (1) common random numbers before and after the bump; (2) the likelihood ratio method, differentiating the density instead of the payoff; (3) payoff smoothing, e.g. a call spread for a digital; (4) for discretely monitored barriers, Brownian-bridge or Broadie-Glasserman-Kou corrections; (5) pathwise or AAD Greeks for smooth payoffs.

中文：有限差分遇到不连续 payoff 时，实际上是在对指示函数求导：每条路径的导数几乎处处为 0，只在跨过 K 的路径上跳变，所以估计器方差极大。解决办法：① **Common random numbers**：bump 前后用同一组随机数；② **Likelihood ratio method**：对密度求导而不是对 payoff 求导；③ **Payoff smoothing**：比如用 call spread 近似 digital；④ **离散监测修正**：Brownian bridge 或 Broadie-Glasserman-Kou；⑤ **Pathwise / AAD**：适用于光滑 payoff。

10. 对冲与 P&L / Hedging and P&L (Q76–Q81)

Q76. Write the Taylor expansion of option P&L. How is it used in P&L explain?

$$\Delta V \approx \Delta S + \frac{1}{2} \Gamma (\Delta S)^2 + \text{Vega} \Delta \sigma + \Theta \Delta t + \rho \Delta r + \text{Vanna} \Delta S \Delta \sigma + \frac{1}{2} \text{Volga} (\Delta \sigma)^2$$

EN: P&L explain attributes the day's actual P&L to each Greek; what's left is unexplained P&L. A large residual points to a missing risk factor, higher-order terms on a big-move day, model or data problems, or new trades and amendments. Under FRTB, the P&L attribution test compares front-office hypothetical P&L with the risk model's risk-theoretical P&L to decide whether a desk may use the internal models approach.

中文：P&L explain 把当天的实际 P&L 分解到各个 Greek 上，剩下的是 **unexplained P&L**。剩余很大说明可能遗漏了风险因子、高阶项不可忽略（大幅变动日）、模型或数据有问题，或者有新交易、交易修改。FRTB 下的 **P&L attribution test** 就是比较前台 hypothetical P&L 与风险模型的 risk-theoretical P&L，决定交易台能否使用内部模型法（IMA），对 risk 岗很加分。

Q77. What is the P&L of a delta-hedged option position?

EN: With continuous hedging, each instant earns the difference between realised and implied variance, weighted by dollar gamma. So you can lose even if realised vol averages above implied, if the big moves happen when gamma is small, e.g. spot far from the strike:

that's path dependency. Hedging at implied vol gives path-dependent P&L; hedging at realised vol gives a known final P&L but a noisy path (Ahmad & Wilmott 2005, a CQF classic).

$$d\{\text{P\&L}\} = \frac{1}{2} \Gamma S^2 \left(\sigma_{\{\text{realised}\}}^2 - \sigma_{\{\text{implied}\}}^2 \right) dt$$

中文：对冲之后赚的是实际波动率和隐含波动率之差，按 **dollar gamma** 加权。含义有二：① 就算整体实际波动率高于隐含，如果大幅变动发生在 gamma 很小的时候（标的远离行权价），仍可能亏钱，这就是 **path dependency**；② 按 implied vol 对冲，P&L 路径依赖；按 realised vol 对冲，最终 P&L 确定但路径波动大。文献是 Ahmad & Wilmott (2005)，CQF 的经典内容。

Q78. What is gamma scalping?

EN: Hold long gamma, e.g. a long straddle, and keep delta-hedging: sell as spot rises and delta grows, buy as spot falls. Each round trip locks in a small profit, paid for by daily theta. You make money if realised vol exceeds implied; it's the practical version of Q77.

中文：持有 long gamma（比如 long straddle），然后反复 delta 对冲：标的上涨、delta 变大时卖出，下跌时买回。每次往返锁定一点利润，代价是每天付出的 theta。标的实际波动大于 implied 时赚钱，否则亏钱，这就是 Q77 公式的实操版本。

Q79. How often should you rebalance a delta hedge?

EN: It's a trade-off: hedging error falls roughly like $1/\sqrt{N}$ with N rebalances, while transaction costs grow linearly in N . In practice: time-based hedging (e.g. at each close); band-based hedging, rebalancing only when delta drifts past a threshold (Whalley-Wilmott optimal bands depend on gamma and costs); or a Leland adjustment, pricing costs in as a higher effective vol. Higher gamma (ATM, near expiry) means more frequent hedging.

中文：这是一个权衡。对冲越频繁，误差越小（大致按 $1/\sqrt{N}$ 下降），但交易成本按 N 线性增长。实务做法：**定时对冲**（比如每天收盘）；**定带宽对冲**（delta 偏离超过阈值才调，对应 Whalley-Wilmott 最优带宽）；**Leland 调整**（把交易成本折算成更高的有效波动率）。Gamma 越大（ATM、临近到期），需要对冲得越频繁。

Q80. You sold an option at 22% implied vol; realised vol turns out to be 18%. Did you make money?

EN: With continuous delta hedging and nothing else going on, yes in expectation: you collected rent for gamma at 22% and only paid out 18%. Caveats: (1) path dependency: you can still lose if the realised moves hit when gamma is largest; (2) mark-to-market: if implied vol rose along the way, your short vega shows a loss; (3) hedging costs and discrete-hedging error.

中文：如果持续 delta 对冲且没有其他噪声，**期望上赚钱：**你按 22% 收了 gamma 的租金，实际只用掉 18%。但有三个注意点：① **路径依赖：**实际波动集中在 gamma 最大的时候仍可能亏；② **隐含波动率的按市值计价：**中途 implied vol 上升，short vega 头寸会出现浮亏；③ 对冲成本和离散对冲误差。

Q81. Why does a market maker delta hedge rather than take a view?

EN: A market maker earns the bid-offer spread and the volatility risk premium, not directional bets. Delta hedging turns directional risk into vol risk, which is easier to price and manage and uses less capital. Remaining gamma and vega are managed with limits or netted against other client flow.

中文：做市商的收入来自 bid-offer 价差和 volatility risk premium，不来自方向性押注。Delta 对冲把方向风险转换成更容易定价和管理的波动率风险，资本占用也更低。剩余的 gamma 和 vega 用限额管理，或者和其他客户的流量相互抵消。

11. 快速心算、Parity 与速算方法 / Quick maths, parity and trader shortcuts (Q82-Q91)

Q82. Price a 1-year ATM call in your head: $S = 100$, $\sigma = 20\%$, $r = 0$.

EN: Use the Brenner-Subrahmanyam approximation $C \approx 0.4 \cdot \sigma \cdot S \cdot \sqrt{T} = 0.4 \times 0.2 \times 100 \times 1 = 8$ (exact: 7.97). A straddle is about twice that, $0.8 \cdot \sigma \cdot S \cdot \sqrt{T}$, which equals the expected absolute move since $E|Z| = \sqrt{2/\pi} \approx 0.8$. So a straddle price is roughly the market's implied expected move. It only holds near the ATM forward.

$$C_{\text{ATM}} \approx \frac{1}{\sqrt{2\pi}} \sigma S \sqrt{T} \approx 0.4 \sigma S \sqrt{T}$$

中文：代入得 8，精确值 7.97。Straddle 约为两倍，即 $0.8\sigma S\sqrt{T}$ ，正好是标的期望绝对变动幅度 ($E|Z| = \sqrt{2/\pi} \approx 0.8$)。所以有一个好用的说法：**straddle 价格 $\approx$ 市场隐含的预期变动幅度**。这个近似只在 ATM forward 附近成立。

Follow-up / 追问：Why is the ATM price proportional to $\sqrt{T}$?

EN: Variance of independent returns adds linearly in time, so the standard deviation of S_T grows like $\sigma\sqrt{T}$. An ATM option is worth roughly the expected positive part of a normal with that standard deviation, $s/\sqrt{2\pi}$ with $s = F\sigma\sqrt{T}$. Price depends on σ and T only through $\sigma\sqrt{T}$, so it's almost linear in σ (volga ≈ 0 , Q67) and grows like $\sqrt{T}$, which gives $\Theta \propto 1/\sqrt{T}$ (Q13) and Vega $\propto \sqrt{T}$ (Q22). It breaks down when $\sigma\sqrt{T}$ is large, since price can never exceed S (Q27).

$$\mathbb{E}[\max(X, 0)] = \int_0^\infty x \frac{1}{s\sqrt{2\pi}} e^{-x^2/2s^2} dx = \frac{s}{\sqrt{2\pi}} \approx 0.4 F \sigma \sqrt{T}$$

中文：根源在于独立增量的方差随时间线性累加：T年后对数收益率的方差是 $\sigma^2 T$ ，标准差是 $\sigma\sqrt{T}$ ，分布的宽度按 $\sqrt{T}$ 增长。ATM 期权没有内在价值，价格完全来自这个宽度；call 就是只保留正的那一半。严格版本：K = F 时 $C = e^{(-rT)}F[2N(\sigma\sqrt{T}/2) - 1]$ ，对 N 在 0 处一阶展开即得上式。只在 $\sigma\sqrt{T}$ 较小时成立，比如 $\sigma = 80\%$ 、 $T = 5$ 年时增长慢于 $\sqrt{T}$ ，并且永远不超过 S。

Q83. Annual vol is 16%. What's the typical daily move?

EN: Daily vol = annual vol / $\sqrt{252} \approx$ annual / 16, so about 1%. That's the "divide by 16" rule. Conversely, VIX at 32 implies roughly 2% daily moves in the S&P.

中文： $\sigma_{\text{daily}} = \sigma_{\text{annual}} / \sqrt{252} \approx \sigma/16$ ，所以约 1%。这是最常用的“除以 16”法则。反过来，VIX = 32 意味着市场预期标普每天波动约 2%。

Q84. You're long an ATM straddle, delta-hedged. What daily move do you need to break even on theta?

EN: About implied vol divided by 16. From $\Theta \approx -\frac{1}{2}\sigma^2 S^2 \Gamma$, the day's gamma P&L $\frac{1}{2}\Gamma(\delta S)^2$ equals theta when $(\delta S/S)^2 = \sigma^2 \delta t$. At 20% implied you need roughly 1.25% a day.

中文：每天需要的波动大约是隐含波动率除以 16。由 $\Theta \approx -\frac{1}{2}\sigma^2 S^2 \Gamma$ ，当日 gamma 收益 $\frac{1}{2}\Gamma(\delta S)^2$ 要等于 theta，需要 $(\delta S/S)^2 = \sigma^2 \delta t$ 。隐含波动率 20% 时，每天需要约 1.25% 才能保本。

Q85. S = 100, r = 5%, 1-year ATM call trades at 10.45. Where should the put trade?

EN: By parity, $P = C - S + Ke^{(-rT)} = 10.45 - 100 + 95.12 = 5.57$. If the put trades elsewhere, either there's an arbitrage or dividends or borrow costs differ, which is usually the point the interviewer wants you to raise.

中文：由平价关系， $P = C - S + Ke^{(-rT)} = 10.45 - 100 + 95.12 = 5.57$ 。如果市场上的 put 不是这个价，要么有套利机会，要么分红、借券费用有差异，面试官往往就是想听你说出这一点。

Q86. A 100/110 call spread costs 6. What does the market imply?

EN: The maximum payoff is 10, so the price must lie between 0 and $10e^{(-rT)}$. Price / width = 0.6, roughly the discounted risk-neutral probability of finishing above about 105. As the width shrinks, call spread / width becomes a digital (Q71). Anyone selling it to you above 10 is giving you an arbitrage.

中文：最大 payoff 是 10，所以价格必须在 0 到 $10e^{(-rT)}$ 之间。价格 / 宽度 = 0.6，大约是风险中性下到期高于 105 附近的（折现）概率。宽度趋近 0 时，call spread 除以宽度就是 digital (Q71)。如果有人以超过 10 的价格卖给你，就是套利。

Q87. Give the Greek signs for: short straddle, long butterfly (at the body), long calendar.

Position / 头寸	Delta	Gamma	Vega	Theta
Short straddle (ATM)	≈ 0	-	-	+
Long butterfly (spot at the body)	≈ 0	-	-	+
Long calendar (short near, long far)	≈ 0	-	+	+
Long risk reversal (long call, short put)	+	≈ 0	≈ 0	≈ 0

EN: Within one maturity, gamma and theta always have opposite signs. The calendar is the classic case where gamma and vega have opposite signs.

中文：同一期限内 gamma 和 theta 的符号总是相反。Calendar 是典型的例外，它的 gamma 和 vega 符号相反。

Q88. How do you use put-call parity to price one option from another?

$$C - P = S - PV(K) - PV(\text{div}), \quad PV(K) \approx K(1 - rT)$$

EN: Three uses. (1) Given the call, the same-strike put follows directly: with $S = 100$, $K = 100$, $r = 5\%$, $T = 1$ and $C = 10.45$, $P = 10.45 - 100 + 95.12 = 5.57$. (2) Price ITM options through OTM ones: an ITM call = intrinsic value + the same-strike OTM put (plus a small carry term). With $r = 0$, the 90-strike call = $10 + P(90) = 13.59$. Traders always estimate the OTM side because it is pure time value, so the estimate has the least error. (3) For a spot-ATM strike, split the carry term in half: the ATM-forward value is about $0.4\sigma\sqrt{T}$ for both, and the call and put differ by $S - PV(K)$. That gives $C \approx 8 + 2.44 = 10.44$ (exact 10.45) and $P \approx 8 - 2.44 = 5.56$ (exact 5.57).

$$C \approx 0.4\sigma\sqrt{T} + \frac{1}{2}[S - PV(K)], \quad P \approx 0.4\sigma\sqrt{T} - \frac{1}{2}[S - PV(K)]$$

中文：三种用法。① 已知 call 求同行权价的 put（或反过来）： $P = 10.45 - 100 + 95.12 = 5.57$ 。② ITM 期权用 OTM 期权来估：ITM call = 内在价值 + 同行权价的 OTM put（再加一个很小的 carry 项）。 $r = 0$ 时 $K = 90$ 的 call = $10 + P(90) = 13.59$ 。交易员总是先估 OTM 那一边，因为它的价格就是纯时间价值，估算误差最小。③ Spot ATM 期权“一半一半”拆解：ATM forward 的 call 和 put 价格相等，都约为 $0.4\sigma\sqrt{T}$ ；行权价放在现价时，两者差额是 carry 项 $S - PV(K)$ ，各分一半。

Q89. How can you back out the forward, dividends or interest rates from option prices?

EN: (1) Implied forward: at one strike, $C - P = PV(F - K)$, so $F = K + (C - P) \cdot e^{rT}$. This forward already includes dividends and borrow costs; if it disagrees with the one computed from spot, the market's implied dividend or borrow cost differs from your assumption. (2) Box spread: $[C(K1) - P(K1)] - [C(K2) - P(K2)] = PV(K2 - K1)$, which

reveals the market's implied risk-free rate. (3) Conversion / reversal: long stock + long put + short call is riskless and should earn the risk-free rate; any deviation is an arbitrage or a funding opportunity.

中文：① **Implied forward**：同一行权价下 $C - P = PV(F - K)$ ，所以 $F = K + (C - P) \cdot e^{(rT)}$ 。这个 forward 已经包含分红和借券成本；如果和用现货算出来的对不上，说明市场隐含的分红或借券费用与你的假设不同。② **Box spread**：由此可以直接读出市场隐含的无风险利率。③ **Conversion / reversal**：long stock + long put + short call 是无风险组合，锁定的收益应当等于无风险利率，偏离了就是套利或 funding 机会。

Q90. What price bounds and approximations follow directly from payoffs?

Structure / 结构	Constraint from the payoff / payoff 给出的约束	Shortcut / 速算
Call	$\max(S - PV(K), 0) \leq C \leq S$	Deep ITM / 深度 ITM : $C \approx S - PV(K)$; deep OTM / 深度 OTM : $C \approx 0$
Call spread K1/K2	$0 \leq \text{price} \leq PV(K2 - K1)$	Price / width $\approx$ risk-neutral probability of finishing above the midpoint / 到期高于中点的风险中性概率
Digital	Limit of call spread / width as width $\rightarrow 0$ / 宽度趋于 0 的 call spread / 宽度	$\approx e^{(-rT)} \cdot N(d2)$; roughly the delta / 粗略可用 delta
Butterfly	Price ≥ 0 / 价格 ≥ 0	Price / width ² $\approx$ risk-neutral density / 风险中性概率密度
Straddle	$C + P$	ATM $\approx 0.8\sigma S\sqrt{T} \approx$ implied expected move / 市场隐含的预期变动
Same-strike C - P / 同行权价 C - P	Equals a forward / 等于一份 forward	$S - PV(K)$, independent of vol / 与波动率无关

EN: Any violation is an arbitrage: call prices must be decreasing and convex in strike, a call spread can't exceed $PV(\text{width})$, and a butterfly can't be negative.

中文：任何一条被违反都存在套利：call 价格对行权价必须单调递减且凸；call spread 不能超过 $PV(\text{宽度})$ ；butterfly 不能为负。

Q91. What quick rules do traders use to price options and Greeks in their head?

Quantity / 要算的量	Shortcut / 速算公式	Example (S = 100, σ = 20%, T = 1, r = 0; BS in brackets)
ATM option / ATM 期权	$0.4 \times \sigma \times S \times \sqrt{T}$	8.0 (7.97); 3 months / 3 个月 4.0 (3.99)
ATM straddle	$0.8 \times \sigma \times S \times \sqrt{T}$	16 = implied expected absolute move / 隐含预期绝对变动
Vol from straddle / 由 straddle 反推 vol	$\sigma \approx \text{straddle} / (0.8 \cdot S \cdot \sqrt{T})$	16 $\rightarrow$ 20%
Daily / weekly / monthly move / 日、周、月波动	$\sigma/16, \sigma/7.2, \sigma/3.5$	20 vol $\approx$ 1.25% a day / 每天
ATM vega per vol point / 每个 vol 点	ATM price $\div$ vol in points / ATM 价格 $\div$ vol 点数	8 / 20 = 0.40 (0.397)
ATM theta per day / 每天	ATM price $\div$ (2 $\times$ days left) / ATM 价格 $\div$ (2 $\times$ 剩余天数)	8 / 730 = 0.011 (0.0109)
ATM gamma	$0.4 / (S \cdot \sigma \cdot \sqrt{T})$	0.02
1 standard deviation OTM / 1 个标准差 OTM	About 1/4 to 1/5 of ATM / 约 ATM 的 1/4-1/5	1.84
2 standard deviations OTM / 2 个标准差 OTM	About 2-3% of ATM / 约 ATM 的 2-3%	0.21
Delta vs distance / delta 与 OTM 程度	$25\Delta \approx 0.67$ sd from forward; $16\Delta \approx 1$ sd; $2.5\Delta \approx 2$ sd	Strike = $F \cdot e^{(\pm z\sigma\sqrt{T})}$
Straddle daily breakeven / straddle 日保本波动	Implied vol $\div$ 16 / 隐含 vol $\div$ 16	20 vol $\rightarrow$ about 1.25% a day
Forward vol	$\sigma_{\text{fwd}}^2 = (\sigma_2^2 T_2 - \sigma_1^2 T_1) / (T_2 - T_1)$	1Y 20%, 2Y 22% $\rightarrow$ 1Y-forward 1Y vol $\approx$ 23.8%
Earnings-implied move / 财报隐含单日变动	Event variance from the term structure, then $\approx 0.8 \times \sigma_{\text{event}} / \sqrt{252}$ / 先求事件方差再乘 0.8	Judge whether the event is rich / 判断事件定价是否偏贵

EN (why they work):

- **$0.4\sigma S\sqrt{T}$** : an ATM option is the expected positive part of a distribution with standard deviation $s = F\sigma\sqrt{T}$, and for a normal that is $s/\sqrt{2\pi} \approx 0.4s$.

- **Vega = price ÷ vol**: the ATM price is nearly linear in σ (volga ≈ 0), so its derivative is the price divided by σ .
- **Theta = price ÷ (2 × days)**: price $\propto \sqrt{\tau}$, so $dC/d\tau = C/(2\tau)$.
- **OTM shortcut**: price $\approx \sigma S\sqrt{T} \times [\phi(z) - z(1 - N(z))]$ with z standard deviations from the forward; the factor is about 0.083 at $z = 1$ and 0.0085 at $z = 2$. Real prices sit slightly higher because the lognormal distribution has a longer right tail.
- **Where they apply**: near the ATM forward with small $\sigma\sqrt{T}$. For long maturities, high vol, far strikes, or when skew matters, go back to BS and the vol surface.

中文（为什么成立）：

- **$0.4\sigma S\sqrt{T}$** ：ATM 期权等于“分布正半边的期望”。对标准差为 s 的正态分布，这个期望是 $s/\sqrt{2\pi} \approx 0.4s$ ，而 $s = F\sigma\sqrt{T}$ 。
- **Vega = 价格 ÷ vol**：ATM 价格关于 σ 近似线性（volga ≈ 0 ），所以对 σ 的导数就等于价格除以 σ 。
- **Theta = 价格 ÷ (2 × 剩余天数)**：价格 $\propto \sqrt{\tau}$ ，求导得 $dC/d\tau = C/(2\tau)$ 。
- **OTM 速算**：价格 $\approx \sigma S\sqrt{T} \times [\phi(z) - z(1 - N(z))]$ ， z 是离 forward 多少个标准差； $z = 1$ 时系数约 0.083， $z = 2$ 时约 0.0085。实际价格略高，因为对数正态分布右尾更长。
- **适用范围**：ATM forward 附近、 $\sigma\sqrt{T}$ 较小时最准。长期限、高波动、行权价离 forward 很远，或者需要考虑 skew 时，要回到 BS 和 vol surface。

12. 风险与估值实务 / Risk and valuation practice (Q92-Q104)

EN: This section plays to your valuation and risk background; weave in concrete processes from Enfusion and MUFG when you answer.

这一节最能发挥你 valuation 和 risk 的背景。回答时可以穿插在 Enfusion、MUFG 做过的具体流程。

Q92. Why is delta-normal VaR inadequate for an options book? What do you use instead?

EN: Delta-normal VaR treats P&L as linear in the risk factors and ignores gamma and vega. For a short-gamma book the P&L distribution is left-skewed, so it badly understates tail losses. Better, in increasing accuracy: (1) delta-gamma approximations (Cornish-Fisher or delta-gamma-normal); (2) historical or Monte Carlo full revaluation; (3) spot-vol grid interpolation, a good speed/accuracy compromise. Vol must be a risk factor, keeping its negative correlation with spot. The same applies to Expected Shortfall under FRTB.

中文：Delta-normal VaR 把 P&L 当成风险因子的线性函数，忽略了 gamma 和 vega。Short gamma 组合的 P&L 分布左偏，这样算会严重低估尾部损失。替代方法按精度递增：① **Delta-gamma**（Cornish-Fisher 或 delta-gamma-normal）；② **Historical 或 Monte Carlo 的 full**

revaluation；③ **Spot-vol 网格插值**。还要把波动率作为风险因子，保留 spot 与 vol 的负相关。FRTB 下的 ES 同样适用。

Q93. How would you perform IPV (independent price verification) on an options book?

EN: (1) Collect independent data: Totem consensus, exchange settlements, broker quotes. (2) Compare vol surfaces bucketed by maturity and delta, express differences in vol points, then multiply by vega to get P&L impact. (3) Set thresholds; breaches trigger a challenge to the trader's marks and escalation. (4) Hold reserves: bid-offer, model reserves for unobservable inputs, concentration. (5) Assign IFRS 13 fair-value levels; long-dated or deep OTM wings are often Level 3.

中文：① **收集独立数据**：Totem 共识价、交易所结算价、broker quote；② **比对波动率曲面**：按期限和 delta 分桶，以波动率表示差异，再乘以 vega 换算成 P&L 影响；③ **设阈值**：超过就 challenge 交易员的 mark 并升级；④ **计提储备**：bid-offer、model reserve、concentration reserve；⑤ **公允价值层级**：IFRS 13 的 Level 2 还是 Level 3，长期限或深度 OTM 的 wings 往往是 Level 3。

Follow-up / 追问：What IPV thresholds are typical for each asset class?

EN: There is no universal standard; every bank and fund administrator sets thresholds in its own valuation policy, and they aren't public. The ranges below are typical industry orders of magnitude from general practice, not a published source, so treat them as rough guides. In an interview, the logic of how thresholds are set matters more than the numbers.

中文：没有统一的市场标准。每家银行、基金管理人都在自己的 valuation policy 里定阈值，而且不公开。下表是业内常见的量级，来自行业惯例，没有公开出处，只能当大致参考。面试时说出“阈值怎么定”的逻辑，比背数字更重要。

Asset class / 资产类别	IPV source / 数据源	Typical tolerance (approximate) / 常见范围 (大致)
Listed equities, futures / 上市股票、期货	Exchange close / 交易所收盘价	~0; check stale prices and corporate actions / 基本为 0，查 stale price 和 corporate action
G10 government bonds / G10 国债	Brokers, BVAL, Tradeweb	~0.1-0.25 price points, or 1-3bp in yield
IG corporate bonds / IG 公司债	Brokers, TRACE, pricing services	~0.25-1 point
HY / EM bonds	Same, fewer quotes / 报价更少	~1-3 points; distressed 3-5+

Asset class / 资产类别	IPV source / 数据源	Typical tolerance (approximate) / 常见范围 (大致)
CDS	Consensus data (e.g. Markit)	IG ~2-5bp; HY ~10-25bp or 5-10% of spread
G10 interest rate swaps / G10 利率互换	Broker screens, CCP curves	Liquid tenors ~0.5-1bp; long or illiquid 1-3bp
EM interest rate swaps / EM 利率互换	Brokers	~3-10bp
FX spot and forwards	WM/Reuters fix, broker screens	A few pips
FX option vol, G10	Brokers, Totem	ATM ~0.25-0.5 vol pts; RR/BF ~0.25-0.5
FX option vol, EM	Brokers	~0.5-1.5 vol pts
Equity index option vol / 股指期权 vol	Totem, exchanges	ATM ~0.5-1 vol pt; wings 1-3
Single-stock option vol / 个股期权 vol	Totem	~1-2 vol pts; wings wider
Swaption / cap vol	Totem, brokers	ATM ~1-3bp normal vol; wings wider
Commodity futures and vol / 商品	Exchanges; Totem	Futures ~0; vol ~1-2 vol pts
Level 3 / structured products / 结构化产品	Models, consensus, counterparties	Usually no pass/fail threshold; hold a model reserve instead / 通常不设阈值, 改为计提 model reserve

EN (how thresholds are set):

1. **Tie to bid-offer:** a difference within half the bid-offer passes, so less liquid markets get wider thresholds. This is the most common and easiest to defend.
2. **Tie to consensus dispersion:** challenge when the desk's mark is more than 1-2 standard deviations from the Totem submissions.
3. **Materiality by P&L impact:** price difference $\times$ sensitivity (DV01, vega) gives the P&L impact, compared with a money threshold per desk or fund (an amount, or a few bp of NAV). Technical breaches that aren't material are documented, not adjusted.

4. **Tiered escalation:** the analyst checks sources first, then challenges the trader, then escalates to the Valuation Committee, which decides between adjusting the mark and taking a reserve.
5. **Regulatory link:** banks follow EBA prudent valuation (AVAs at a 90% confidence exit price); fair value levels follow IFRS 13 / ASC 820.

Fund administration add-on: price validation also uses day-over-day tolerance checks (e.g. equities moving more than about 5-10% get reviewed), stale-price checks, and NAV-level materiality thresholds.

中文（阈值怎么定）：

1. **与 bid-offer 挂钩：**差异在买卖价差的一半以内就算通过，流动性越差阈值越宽。这是最常见也最好解释的依据。
2. **与共识分散度挂钩：**比如超过 Totem 提交价标准差的 1-2 倍就要 challenge。
3. **按 P&L 影响设重要性门槛：**价格差异 $\times$ 敏感度 (DV01、vega) 得出 P&L 影响，再和按交易平台或基金设定的金额门槛比较。技术上超阈值但影响不重要的，只记录不调整。
4. **分级升级：**分析员核对来源 $\rightarrow$ challenge 交易员 $\rightarrow$ 升级到 Valuation Committee，决定调整 mark 还是计提 reserve。
5. **与监管框架联系：**银行有 EBA Prudent Valuation（按 90% 置信度退出价计提 AVA）；公允价值层级按 IFRS 13 / ASC 820。

基金行业补充：基金管理人的价格验证通常还有日间价格变动容差（比如股票单日变动超过约 5-10% 要核查）、stale price 检查，以及 NAV 层面的重要性门槛。

Q94. Local vol and stochastic vol both fit today's vanilla surface. Why might they price a barrier differently?

EN: They match today's marginal distributions but imply different dynamics. Local vol's forward skew flattens over time and it effectively assumes perfect spot-vol correlation; stochastic vol keeps future skew alive. Barriers and cliquets depend on the future smile, so their prices differ. That gap is model risk, often reserved using the range across candidate models; local-stochastic vol (LSV) is the usual compromise.

中文：两者对今天的边际分布拟合得一样，但动态不同。Local vol 隐含的 forward skew 会随时间变平，并且假设 spot 与 vol 完全相关；stochastic vol 保留了未来的 skew。Barrier 和 cliquet 对未来 smile 的动态敏感，所以价格不同。差异就是 **model risk**，可以用几个模型的价格区间计提 model reserve。折衷方案是 local-stochastic vol (LSV)。

Q95. How do you stress test an options portfolio?

EN: The core is a spot-vol matrix, e.g. spot $\pm 20\%$ by vol ± 10 points, with full revaluation in every cell. Add historical scenarios (1987, 2008, March 2020), vol-surface shape shocks (steeper skew, inverted term structure), correlation going to 1 for dispersion and basket

books, and liquidity shocks (much wider bid-offer). The goal is to find nonlinear losses Greeks can't see, e.g. a short OTM put that only explodes after a 15% fall.

中文：最常用的是 spot-vol 矩阵：比如 spot $\pm 20\%$ 、vol ± 10 个点，每个格子 full revaluation。再加：**历史情景**（1987、2008、2020 年 3 月）、**波动率曲面形状冲击**（skew 变陡、期限结构倒挂）、**相关性趋向 1**（对 dispersion 和 basket 尤其重要）、**流动性冲击**（bid-offer 大幅加宽）。重点是找出 Greeks 看不到的非线性损失。

Q96. What is FRTB SA, and how does it capture option risk?

EN: FRTB (Fundamental Review of the Trading Book) is the Basel market-risk framework finalised in 2019, replacing Basel 2.5's VaR rules. The Standardised Approach is mandatory for all banks and is the floor and fallback for the Internal Models Approach (Expected Shortfall, subject to P&L attribution and backtesting). SA capital = the sensitivities-based method (delta, vega by maturity bucket, and curvature, which captures gamma by full revaluation under large up/down shocks minus the delta part) + the default risk charge + the residual risk add-on (1% of notional for exotic underlyings, 0.1% for other residual risks). So short gamma and exotic positions are penalised twice, via curvature and RRAO.

$$\text{CVR}_k = -\min\left(V(x_k^{+RW}) - V(x_k) - RW \cdot \Delta_k, V(x_k^{-RW}) - V(x_k) + RW \cdot \Delta_k\right)$$

中文：FRTB 是巴塞尔委员会 2019 年定稿的新版市场风险框架。SA（标准法）所有银行都必须算，也是 IMA 的兜底和对照。SA 资本 = **SBM**（delta、按期限分桶的 vega、curvature）+ **DRC**（违约跳跃风险）+ **RRAO**（奇异标的按名义 1%，其他残余风险按 0.1%）。SBM 的敏感度乘以监管风险权重，先桶内、再跨桶加总，相关系数按低、中、高三种情景各算一次取最大，共 7 类风险：GIRR、三类 CSR、股票、商品、外汇。Curvature 本质上是监管版的 gamma 风险。实施时间：英国 PRA 计划 SA 于 2027 年 1 月 1 日生效、IMA 推迟到 2028 年 1 月 1 日（[PRA CP9/26](#)）；欧盟委员会 2025 年 6 月提议推迟到 2027 年 1 月 1 日（[European Commission](#)），面试前再核实一下。

Q97. How are Greeks computed in production, and what can go wrong?

EN: Analytic formulas for vanillas; bump-and-reprice as the general method (choose bump sizes, e.g. 1% spot, 1 vol point, and one-sided vs central differences); AAD for all sensitivities in one pass on large Monte Carlo books. What goes wrong: bump size (too small is numerical noise, too big mixes in higher-order effects); Monte Carlo noise (use common random numbers, Q75); inconsistent conventions between front office and risk (sticky assumptions, units, calendars) breaking P&L explain; and Greeks jumping near barriers and digitals.

中文：三种方法：**解析式**（vanilla）、**Bump and reprice**（最通用，要选步长和单边/中心差分）、**AAD**（一次算出所有敏感度）。常见问题：**步长选择**（太小有数值噪声、太大混入高阶效应）、**Monte Carlo 噪声**（同一组随机数，Q75）、**口径不一致**（前台和风险系统的 sticky 假

设、单位、日历不同，P&L explain 对不上）、**不连续 payoff**（barrier、digital 附近 Greeks 剧烈跳动）。这是典型的跨部门对账问题，也是 valuation 岗的日常。

Q98. Why can you scale VaR by $\sqrt{T}$, and is it the same principle as ATM option pricing?

EN: Yes, same root: variance of independent, identically distributed returns adds linearly in time, so the scale of the P&L distribution grows like $\sigma\sqrt{T}$. An ATM option prices the expected positive part ($\approx 0.4\sigma\sqrt{T}$); 99% VaR takes a tail quantile ($\approx 2.33\sigma\sqrt{T}$); same width, different functional. That's why 10-day VaR = 1-day $\times \sqrt{10}$, and FRTB scales ES by $\sqrt{LH/10}$ across liquidity horizons. It breaks when returns are autocorrelated, vol clusters, tails are fat, the horizon is long enough for drift to matter, or the portfolio is non-linear; for an options book, simulate the target horizon directly.

$$\text{Var}(\sum_{t=1}^T r_t) = T\sigma^2 \quad ; \rightarrow ; \\ \text{VaR}_T \approx z_{\alpha} \sigma \sqrt{T} , S$$

中文：底层原理完全一样，都是“分布宽度 $\times$ 常数”：ATM 期权取正半边期望（系数 0.4），99% VaR 取左尾 1% 分位数（系数 2.33），straddle 取 $E|X|$ （系数 0.8）。 $\sqrt{T}$ 规则成立需要的假设（追问重点）：

1. **收益独立、无自相关**： $\text{Var}(\sum r) = \sigma^2[T + 2\sum(T-k)\rho_k]$ 。正自相关（动量） $\rightarrow \sqrt{T}$ 低估风险；负自相关（均值回归） $\rightarrow$ 高估。
2. **波动率恒定**：现实有波动率聚集（GARCH）。高波动时期波动率会均值回归，直接用 $\sqrt{T}$ 放大会高估长期风险，平静期则相反。
3. **分布形状不变**：正态加总后仍是正态，系数 2.33 不变；但日收益是厚尾的，加总后趋向正态、峰度下降。Danielsson & de Vries 指出极端尾部分位数更接近按 $T^{1/\alpha}$ 缩放。
4. **忽略漂移**：均值按 T 、标准差按 $\sqrt{T}$ ，长限时 $\text{VaR}_T \approx z\sigma\sqrt{T} - \mu T$ 。
5. **头寸静态且线性**：期权 $\text{P\&L} \approx \Delta \cdot \delta S + \frac{1}{2}\Gamma \cdot (\delta S)^2$ ，delta 项按 $\sqrt{T}$ ，gamma 项按 T 缩放且不对称，所以 **short gamma 组合的 VaR 不能用 $\sqrt{T}$ 放大**，必须按目标期限直接模拟或 full revaluation。

VaR 回测 / VaR backtesting (Q99–Q104)

Q99. What is VaR backtesting, and what P&L do you compare against?

EN: Backtesting checks whether the VaR model's stated confidence level holds in reality. Each day you compare the prior day's VaR with the next day's P&L; a loss larger than VaR is an exception (breach). With 99% one-day VaR you expect about 1% exceptions, i.e. 2.5 in 250 trading days. The P&L matters: hypothetical P&L revalues yesterday's frozen positions at today's prices, so it tests the model cleanly; actual P&L includes intraday trading, new deals, fees and reserves, so it tests the whole desk. Regulators require both. Fees, commissions and valuation adjustments that the VaR model doesn't cover should be excluded, which is why desks talk about “clean” vs “dirty” P&L.

中文：回测就是检验 VaR 模型所说的置信度在现实中是否成立。每天拿前一天的 VaR 和次日的 P&L 比较，亏损超过 VaR 就记一次 **exception (突破)**。99% 单日 VaR 预期突破比例约 1%，即 250 个交易日约 2.5 次。比较用哪种 P&L 很关键：**hypothetical P&L** 是把昨天收盘的头寸冻结、用今天的价格重估，最干净地检验模型；**actual P&L** 包含日内交易、新交易、费用和储备，检验的是整个交易台。监管要求两者都做。VaR 模型不覆盖的手续费、佣金和估值调整要剔除，这就是交易台常说的 clean 与 dirty P&L。

Q100. Explain the Basel traffic-light approach.

EN: Count exceptions of 99% one-day VaR over the last 250 days. Zones are set from the binomial distribution: green if the count could plausibly come from a correct model (cumulative probability below 95%), red if it's almost impossible (at or above 99.99%). Under Basel 2.5 the capital multiplier is 3 plus a plus factor, and capital = max(yesterday's VaR, multiplier $\times$ 60-day average VaR), plus the stressed VaR term. Red means the model is presumed flawed and the supervisor can require action.

Exceptions / 突破次数	Zone / 区域	Cumulative probability (correct model) / 累计概率	Plus factor	Multiplier / 乘数
0-4	Green / 绿	$\leq 89.2\%$	0.00	3.00
5	Yellow / 黄	95.9%	0.40	3.40
6	Yellow	98.6%	0.50	3.50
7	Yellow	99.6%	0.65	3.65
8	Yellow	99.9%	0.75	3.75
9	Yellow	99.97%	0.85	3.85
10+	Red / 红	$\geq 99.99\%$	1.00	4.00

中文：统计过去 250 天中 99% 单日 VaR 的突破次数。区域由二项分布决定：如果模型正确，出现这么多次突破的累计概率低于 95% 为**绿区**（0-4 次），高于或等于 99.99% 为**红区**（10 次及以上），中间为**黄区**。Basel 2.5 下资本乘数 = 3 + plus factor，资本 = max(前一日 VaR，乘数 $\times$ 60 日平均 VaR)，再加 stressed VaR 项。进入红区即推定模型有缺陷，监管可以要求整改。

Q101. Which statistical tests do you use to backtest VaR?

EN: (1) Kupiec's proportion-of-failures (POF) test checks whether the exception rate equals the target p. The likelihood-ratio statistic below is compared with the $\chi^2(1)$ critical value 3.84 at 95%. With 250 days at 99%: 4 exceptions gives LR = 0.77 (accept), 8 gives 7.73 (reject), and 0 gives 5.03 (reject: the model is too conservative, which wastes capital).

(2) Christoffersen's independence test checks whether exceptions cluster, using the probability of an exception the day after an exception; clustering means the model reacts too slowly to volatility changes. (3) Conditional coverage combines both, $LR_{cc} = LR_{POF} + LR_{ind} \sim \chi^2(2)$, critical value 5.99. Power is low with only 250 observations, so tests are complemented by judgement and longer windows.

$$LR_{\{POF\}} = -2 \ln \left[\frac{(1-p)^{T-x} p^x}{(1-p)^T} \right] + 2 \ln \left[\frac{(1-p)^{T-x} p^x}{(1-p)^T} \right] \sim \chi^2_1$$

中文：① **Kupiec POF 检验**：检验突破比例是否等于目标 p ，似然比统计量与 $\chi^2(1)$ 的 95% 临界值 3.84 比较。250 天、99% 下：4 次突破 $LR = 0.77$ （接受），8 次 $LR = 7.73$ （拒绝），**0 次 $LR = 5.03$ （也拒绝）**，说明模型太保守、浪费资本。② **Christoffersen 独立性检验**：看突破是否成簇出现（前一天突破后第二天再突破的概率），成簇说明模型对波动率变化反应太慢，典型是长窗口、等权重的历史模拟法。③ **Conditional coverage** 把两者合并， $LR_{cc} \sim \chi^2(2)$ ，临界值 5.99。只有 250 个观测时检验功效很低，所以还要结合专业判断和更长的窗口。

Q102. How does backtesting work under FRTB?

EN: Two layers. (1) Desk-level eligibility for the internal models approach: each trading desk backtests 99% and 97.5% one-day VaR against both actual and hypothetical P&L over the last 250 days; more than 12 exceptions at 99% or more than 30 at 97.5% (expected 2.5 and 6.25) sends the desk to the standardised approach. (2) The P&L attribution test (PLAT) compares hypothetical P&L (front-office models) with risk-theoretical P&L (the risk model's factors) using the Spearman correlation and the Kolmogorov-Smirnov test; roughly, green needs Spearman ≥ 0.80 and KS ≤ 0.09 , red is Spearman < 0.70 or KS > 0.12 , and amber desks keep IMA with a capital surcharge. Bank-wide 99% VaR backtesting still uses a traffic light that adds a plus factor to the ES capital multiplier.

中文：分两层。① **交易台层面的 IMA 资格**：每个交易台对 99% 和 97.5% 单日 VaR，分别用 actual 和 hypothetical P&L 回测过去 250 天；99% 下超过 12 次、或 97.5% 下超过 30 次突破（预期分别为 2.5 和 6.25 次），该交易台就要改用标准法。② **P&L attribution test (PLAT)**：比较前台模型的 hypothetical P&L 与风险模型风险因子算出的 risk-theoretical P&L，用 Spearman 相关系数和 Kolmogorov-Smirnov 检验。大致标准：绿区 Spearman ≥ 0.80 且 KS ≤ 0.09 ；红区 Spearman < 0.70 或 KS > 0.12 ；黄区可以继续用 IMA，但要加资本。银行整体的 99% VaR 回测仍然用红绿灯，在 ES 资本乘数上加 plus factor。

Q103. You see too many (or zero) exceptions. How do you investigate?

EN: Treat each exception as a case. (1) Check the P&L first: booking errors, late trades, stale or wrong market data, reserve or fee moves that shouldn't be in clean P&L. (2) Attribute the loss: was it a genuine tail move, a risk factor the model doesn't capture (basis, dividends, correlation, jumps), a proxy that behaved differently, or a nonlinearity (options) that delta-based VaR misses? (3) Look at the model: window too short or too

long, equal weighting reacting slowly, missing vol-of-vol or skew. (4) Document and remediate: add risk factors, reduce proxies, change the weighting, or hold a VaR add-on until fixed. Zero exceptions over a year is also a finding: the model is too conservative, which ties up capital and suggests over-hedged assumptions.

中文：每次突破都要当作一个案子来查。① **先查 P&L**：是否有录入错误、晚到的交易、过期或错误的市场数据，或不该进 clean P&L 的储备、费用变动。② **归因**：是真正的尾部行情，还是模型没覆盖的风险因子（基差、分红、相关性、跳跃）、代理因子走势不同，或者 delta 法 VaR 漏掉的非线性（期权）？③ **查模型**：窗口太短或太长、等权重反应太慢、缺少 vol-of-vol 或 skew。④ **记录并整改**：增加风险因子、减少代理、改权重，或在修复前加 VaR add-on。一年零突破也是发现：模型太保守，占用多余资本。

Q104. Can you backtest Expected Shortfall the same way?

EN: Not directly. ES is the average loss beyond VaR, so a single day can't "breach" it the way it breaches VaR, and ES is not elicitable, meaning there is no simple scoring function whose minimiser is ES. Approaches include the Acerbi-Székely tests, which compare realised tail losses with predicted ES, and testing VaR at several quantiles as a proxy for the tail. That is why FRTB sets capital with 97.5% ES but still backtests VaR at 99% and 97.5%.

中文：不能直接用同样的方法。ES 是超过 VaR 部分的平均损失，单日没有“突破 ES”这种二元事件；而且 ES **不具备 elicibility**，没有一个简单的评分函数能让 ES 成为最优解。常见做法：Acerbi-Székely 检验（把实际尾部损失与预测 ES 比较），或者在多个分位数上回测 VaR 来间接检验尾部。这就是为什么 FRTB 用 97.5% ES 计算资本，却仍然在 99% 和 97.5% 上回测 VaR。

13. 深度进阶 / Advanced deep dives (Q105–Q110)

EN: These questions separate candidates who know the formulas from those who understand how the machinery connects. They're common follow-ups in quant, valuation and market-risk interviews at banks and funds.

中文：这几道题区分“会背公式”和“懂整套机制怎么联动”的候选人，是银行和基金 quant、valuation、market risk 面试里常见的深入追问。

Q105. How is a variance swap replicated, and why are the options weighted by $1/K^2$?

$$\int_0^T \sigma_t^2 dt = 2 \int_0^T \frac{dS_t}{S_t} - 2 \ln \frac{S_T}{S_0}, \\ \text{qqquad } K_{\text{var}} = \frac{2e^{rT}}{T} \left[\int_0^T \frac{P(K)}{K^2} dK + \int_0^{\infty} \frac{C(K)}{K^2} dK \right]$$

EN: Apply Itô to $\ln S$: realised variance equals a continuously delta-hedged position (the dS/S term) minus twice a log contract. The log payoff can be built from a static strip of OTM puts and calls weighted by $1/K^2$, plus a forward. Intuition for $1/K^2$: a vanilla's dollar gamma ΓS^2 is concentrated near its strike; weighting strikes by $1/K^2$ makes the strip's

dollar gamma constant in spot, so its delta-hedged P&L $\frac{1}{2}\Gamma S^2(\sigma_r^2 - \sigma_i^2)$ no longer depends on where spot travels (Q77). That's what makes a variance swap a pure volatility bet. Because the low strikes carry more weight and equity skew makes them expensive, the fair variance strike sits above ATM vol; it also means variance swaps lose heavily in crashes, which is why single-name variance is usually capped.

中文：对 $\ln S$ 用 Itô 引理：实现方差 = 连续 delta 对冲的头寸 (dS/S 项) 减去两倍的 log 合约。Log payoff 可以用一组按 $1/K^2$ 加权的 OTM put 和 call、再加一份 forward 静态复制。 **$1/K^2$ 的直觉：**单个 vanilla 的 dollar gamma ΓS^2 集中在它的行权价附近；按 $1/K^2$ 加权后，整组期权的 dollar gamma 对 spot 恒定，所以 delta 对冲的 P&L $\frac{1}{2}\Gamma S^2(\sigma_r^2 - \sigma_i^2)$ 不再取决于 spot 走了什么路径 (Q77)，这才是纯粹的波动率押注。低行权价权重更大，而股票 skew 让它们更贵，所以方差互换的公允行权价高于 ATM vol；也因此方差互换在崩盘时亏得很多，个股方差互换通常会加 cap。

Q106. Explain the SABR model's parameters. What does each control, and what are its known problems?

$$dF_t = \alpha_t F_t^{\beta} dW_t^1, \quad d\alpha_t = \nu \alpha_t dW_t^2, \quad dW_t^1 dW_t^2 = \rho dt$$

EN: α sets the level (ATM vol $\approx \alpha / F^{(1-\beta)}$). β sets the backbone, i.e. how ATM vol moves when the forward moves: $\beta = 1$ is lognormal (ATM vol roughly stays put), $\beta = 0$ is normal (lognormal ATM vol rises as rates fall). ρ sets the skew (spot-vol correlation), and ν (vol of vol) sets the smile curvature, i.e. the wings. In practice β is fixed by market convention and α, ρ, ν are calibrated per expiry to ATM, RR and BF-type quotes. Known problems: Hagan's closed-form approximation can imply negative densities at low strikes and long expiries, which is an arbitrage; negative rates break the F^β term. Fixes include arbitrage-free SABR (solving the PDE), shifted SABR for negative rates, and checking the implied density before using the smile.

中文： α 决定水平 (ATM vol $\approx \alpha / F^{(1-\beta)}$)。 β 决定 backbone，即 forward 变动时 ATM vol 怎么跟着变： $\beta = 1$ 是对数正态 (ATM vol 基本不变)， $\beta = 0$ 是正态 (利率下降时对数正态 ATM vol 上升)。 ρ 决定 skew (spot 与 vol 的相关性)， ν (vol of vol) 决定 smile 的弯曲程度，也就是两翼。实务中 β 按市场惯例固定， α, ρ, ν 按每个期限校准到 ATM、RR、BF 这类报价。已知问题：Hagan 的闭式近似在低行权价、长时期时可能隐含负的概率密度 (即存在套利)；负利率会使 F^β 失效。解决办法：无套利 SABR (解 PDE)、针对负利率的 shifted SABR，以及使用前检查隐含密度。

Q107. How do you price a quanto option, and where does the correlation adjustment come from?

$$F^{\text{quanto}} = F \cdot e^{-\rho \sigma_S \sigma_X T}$$

EN: A quanto pays a foreign asset's value in the domestic currency at a fixed rate, e.g. the Nikkei level paid in USD, 1 dollar per point. Here X is the domestic price of one unit of foreign currency (USD per JPY), σ_X its vol, and ρ the correlation between the asset and X . Under the domestic pricing measure the asset's drift picks up $-\rho\sigma_X\sigma$, so you price with Black-Scholes on the adjusted quanto forward using the asset's vol. Intuition: the seller hedges with the asset in foreign currency and must convert gains at an FX rate that is correlated with those gains; the adjustment is the expected cost of that correlation. For the Nikkei, which tends to rise when the yen weakens ($\rho < 0$ with this convention), the quanto forward sits above the local forward. The trader carries correlation risk (cross-gamma) that can't be hedged with vanillas, and it's typically reserved for in valuation.

中文：Quanto 把外国资产的价值按固定汇率用本币支付，比如日经指数每点付 1 美元。这里 X 是一单位外币的本币价格（每日元的美元价）， σ_X 是它的波动率， ρ 是资产与 X 的相关性。在本币定价测度下，资产的漂移多了一项 $-\rho\sigma_X\sigma$ ，所以用资产自己的波动率、对调整后的 quanto forward 套 Black-Scholes 即可。直觉：卖方用外币计价的资产对冲，赚到的钱要按一个与收益相关的汇率换回本币，这个调整就是该相关性的期望成本。日经往往在日元贬值时上涨（按此汇率定义 $\rho < 0$ ），所以 quanto forward 高于本地 forward。交易员承担无法用 vanilla 对冲的相关性风险（cross-gamma），估值时通常要计提储备。

Q108. What is prudent valuation, and how does it differ from fair value? What is Day-1 P&L?

EN: Fair value (IFRS 13) is the exit price, the mid-market estimate adjusted for bid-offer and other valuation adjustments. Prudent valuation (EU CRR Article 105 and the EBA technical standards) asks for a value you're 90% confident you could exit at, and the difference, the additional valuation adjustments (AVAs), is deducted from CET1 capital. The EBA lists nine AVA categories: market price uncertainty, close-out costs, model risk, unearned credit spreads, investing and funding costs, concentrated positions, future administrative costs, early termination, and operational risk. Smaller portfolios can use a simplified approach. Day-1 P&L: if a trade's price differs from its model value at inception and the valuation relies on significant unobservable (Level 3) inputs, IFRS 9 and 13 require deferring that profit and recognising it as inputs become observable or the trade matures, not booking it upfront.

中文：公允价值（IFRS 13）是退出价，即中间价加上 bid-offer 等估值调整。**审慎估值**（欧盟 CRR 第 105 条和 EBA 技术标准）要求一个有 90% 把握能退出的价格，差额即 **AVA（额外估值调整）**，从 CET1 资本中扣除。EBA 列了九类 AVA：市场价格不确定性、平仓成本、模型风险、未赚取的信用利差、投资与融资成本、集中度、未来管理成本、提前终止、操作风险；规模较小的银行可用简化法。**Day-1 P&L**：如果交易成交价与模型价值不同，而估值依赖重大的不可观察（Level 3）输入，IFRS 9 和 IFRS 13 要求递延这部分利润，等输入变得可观察或交易到期时再确认，不能成交当天入账。

Q109. What makes a volatility surface arbitrage-free, and how does that connect to local volatility?

$$\sigma_{\text{loc}}^2(K,T) = \frac{\frac{\partial C}{\partial T} + (r-q)K \frac{\partial C}{\partial K} + qC}{\frac{1}{2} K^2 \frac{\partial^2 C}{\partial K^2}}$$

EN: Two conditions. No calendar arbitrage: total implied variance $\sigma^2(K,T) \cdot T$ must be non-decreasing in T at fixed moneyness, otherwise a calendar spread is free money. No butterfly arbitrage: call prices must be convex in strike, i.e. the risk-neutral density $\partial^2 C / \partial K^2$ must be non-negative, otherwise a butterfly has negative cost. Dupire's formula turns the surface into local vol: the numerator is the calendar condition and the denominator is the butterfly condition, so any arbitrage shows up as a negative or infinite local variance. That's why surfaces are fitted with arbitrage-free parametrisations (e.g. SVI with constraints) and smoothed before local-vol or exotic pricing. In valuation, an arbitrage in the marked surface is a control failure to fix, not just a modelling detail.

中文：两个条件。① **无 calendar 套利**：同一 moneyness 下，总隐含方差 $\sigma^2(K,T) \cdot T$ 必须随 T 单调不减，否则 calendar spread 就是白捡的钱。② **无 butterfly 套利**：call 价格对行权价必须是凸的，即风险中性密度 $\partial^2 C / \partial K^2$ 非负，否则 butterfly 的成本为负。Dupire 公式把 surface 转成 local vol：分子对应 calendar 条件，分母对应 butterfly 条件，所以任何套利都会表现为负的或无穷大的 local variance。这就是为什么要用无套利的参数化（如加约束的 SVI）拟合 surface、并在 local vol 或奇异期权定价前平滑。在估值岗位上，mark 出来的 surface 有套利是要修正的控制缺陷，而不只是建模细节。

Q110. How does counterparty credit risk (CVA) affect option valuation?

$$\text{CVA} \approx (1-R) \sum_i \text{DF}(t_i) \cdot [\text{EE}(t_i) \cdot \text{PD}(t_{i-1}, t_i)]$$

EN: If you buy an uncollateralised option, the counterparty owes you its value, so your exposure is the option's future value, always non-negative, and CVA is a cost to you. If you sell the option you owe them, so you have no CVA on it, though you may book DVA on your own credit. Wrong-way risk makes it worse: buying crash puts from a bank whose credit collapses in the same crash means the exposure is largest exactly when it's least likely to be paid. Collateral under a CSA cuts exposure to the margin-period-of-risk gap and changes discounting to the collateral (OIS) rate; hedging uncollateralised trades with collateralised hedges creates funding costs (FVA). XVA desks price and hedge these, and valuation teams check they're included in fair value.

中文：买入没有抵押品的期权时，对手欠你期权的价值，你的暴露就是期权未来的价值（永远非负），CVA 是你的成本；卖出期权时是你欠对手，这笔交易没有 CVA，但可能计入自身信用的 DVA。**Wrong-way risk** 会加重问题：向一家在崩盘中自己信用也会崩溃的银行买崩盘保护 put，意味着暴露最大的时候恰恰最难拿到钱。CSA 下的抵押品把暴露降到保证金风险期

（MPoR）内的缺口，并把折现利率改为抵押品利率（OIS）；用有抵押的对冲去对冲无抵押交易会产生融资成本（FVA）。XVA 交易平台负责定价和对冲这些项目，估值团队要检查它们是否已纳入公允价值。